

IIT MATHEMATICS

LIMIT, CONTINUITY & DIFFERENTIABILITY

WORKBOOK



RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
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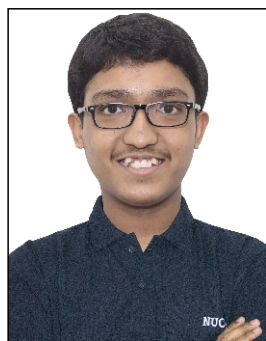


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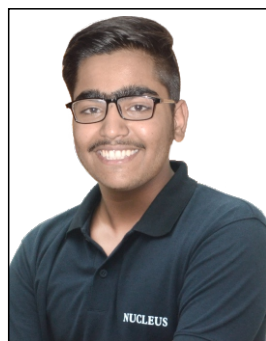


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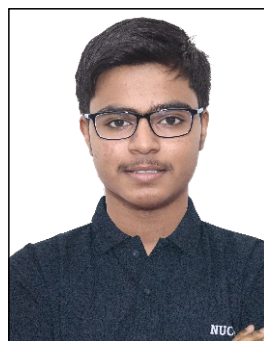
JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



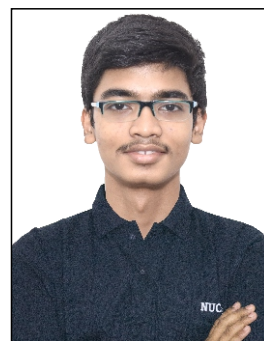
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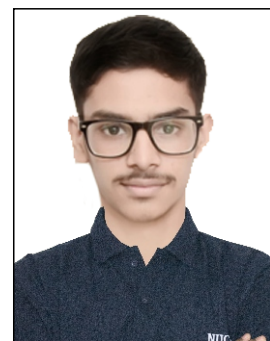
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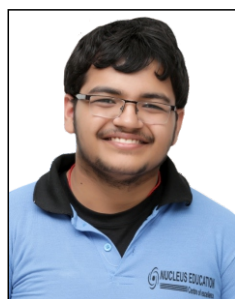


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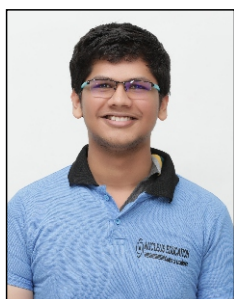


100 Percentile
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2020

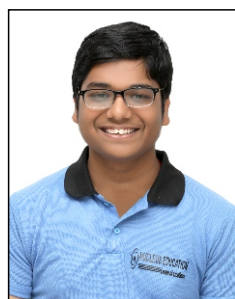
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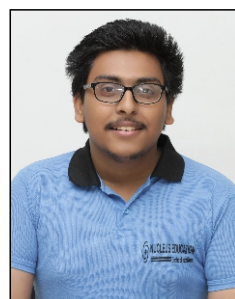
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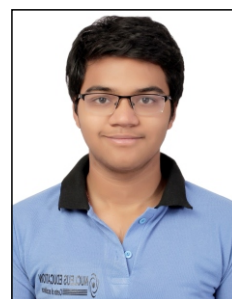
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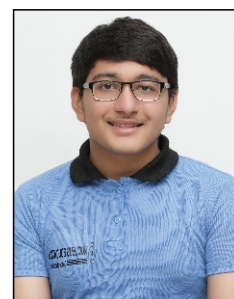
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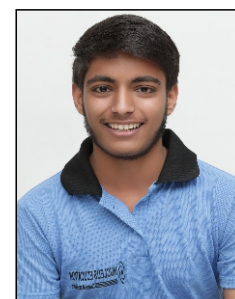
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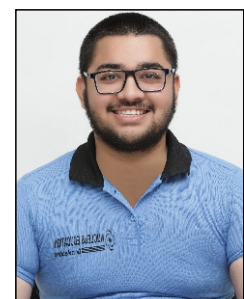
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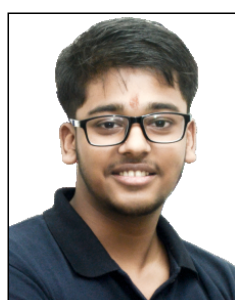
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SATVIK
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MAYANK DUBEY
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HRITHIK
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SHASHANK AGRAWAL
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RAAGHAV RAAJ
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AIR-32
SHREYA PATHAK
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SIDDHANT CHOUDAHRY
Gen. Category - 2018



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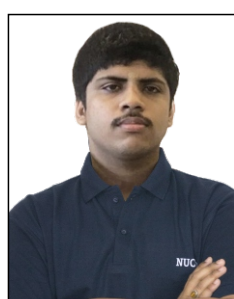
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HIMANSHU GAURAV SINGH
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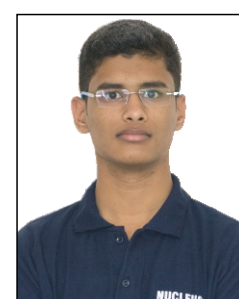
AIR-48
SOHAM MISTRI
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LIMIT

1. INTRODUCTION :

The concept of limit of a function is one of the fundamental ideas that distinguishes calculus from algebra and trigonometry. We use limits to describe the way a function f varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$. Other functions can have values that jump or vary erratically. We also use limits to define tangent lines to graphs of functions. This geometric application leads at once to the important concept of derivative of a function.

2. DEFINITION :

Let $f(x)$ be defined on an open interval about 'a' except possibly at 'a' itself. If $f(x)$ gets arbitrarily close to L (a finite number) for all x sufficiently close to 'a' we say that $f(x)$ approaches the limit L as x approaches 'a' and we write $\lim_{x \rightarrow a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a , equals L ". This implies if we can make the value of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

3. LEFT HAND LIMIT AND RIGHT HAND LIMIT OF A FUNCTION :

The value to which $f(x)$ approaches, as x tends to 'a' from the left hand side ($x \rightarrow a^-$) is called left hand limit of $f(x)$ at $x = a$. Symbolically,

$$\text{LHL} = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h).$$

The value to which $f(x)$ approaches, as x tends to 'a' from the right hand side ($x \rightarrow a^+$) is called right hand limit of $f(x)$ at $x = a$. Symbolically,

$$\text{RHL} = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h).$$

Limit of a function $f(x)$ is said to exist as, $x \rightarrow a$ when $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) =$

Finite quantity.

Example : Graph of $y = f(x)$

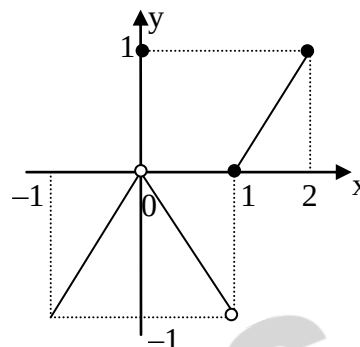


Fig. 1

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{h \rightarrow 0} f(-1 + h) = f(-1^+) = -1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0 - h) = f(0^-) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = f(0^+) = 0$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1 - h) = f(1^-) = -1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1 + h) = f(1^+) = 0$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{h \rightarrow 0} f(2 - h) = f(2^-) = 1$$

$$\lim_{x \rightarrow 0} f(x) = 0 \text{ and } \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

Important note :

In $\lim_{x \rightarrow a} f(x)$, $x \rightarrow a$ necessarily implies

$x \neq a$. That is while evaluating limit at $x = a$, we are not concerned with the value of the function at $x = a$. In fact the function may or may not be defined at $x = a$.

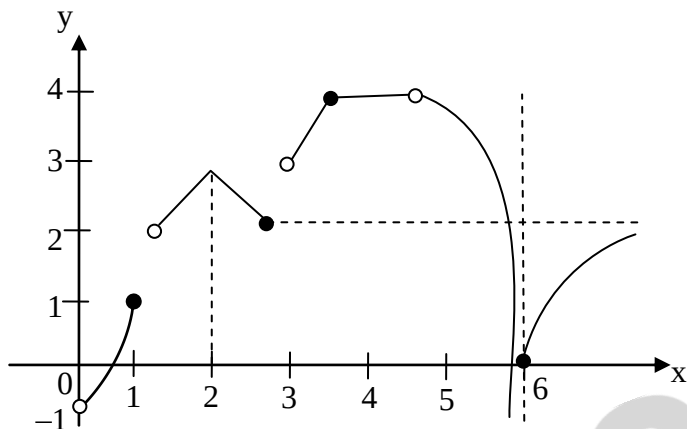
Also it is necessary to note that if $f(x)$ is defined only on one side of ' $x = a$ ', one sided limits are good enough to establish the existence of limits, and if

$f(x)$ is defined on either side of 'a' both sided limits are to be considered.

As in $\lim_{x \rightarrow 1} \cos^{-1} x = 0$, though $f(x)$ is not defined for $x > 1$, even in it's immediate vicinity.

Illustration 1 :

Consider the adjacent graph of $y = f(x)$
Find the following :



- | | |
|---|--|
| (a) $\lim_{x \rightarrow 0^-} f(x)$ | (b) $\lim_{x \rightarrow 0^+} f(x)$ |
| (c) $\lim_{x \rightarrow 1^-} f(x)$ | (d) $\lim_{x \rightarrow 1^+} f(x)$ |
| (e) $\lim_{x \rightarrow 2^-} f(x)$ | (f) $\lim_{x \rightarrow 2^+} f(x)$ |
| (g) $\lim_{x \rightarrow 3^-} f(x)$ | (h) $\lim_{x \rightarrow 3^+} f(x)$ |
| (i) $\lim_{x \rightarrow 4^-} f(x)$ | |
| (j) $\lim_{x \rightarrow 4^+} f(x)$ | (k) $\lim_{x \rightarrow \infty} f(x) = 2$ |
| (l) $\lim_{x \rightarrow 6^-} f(x) = -\infty$ | |

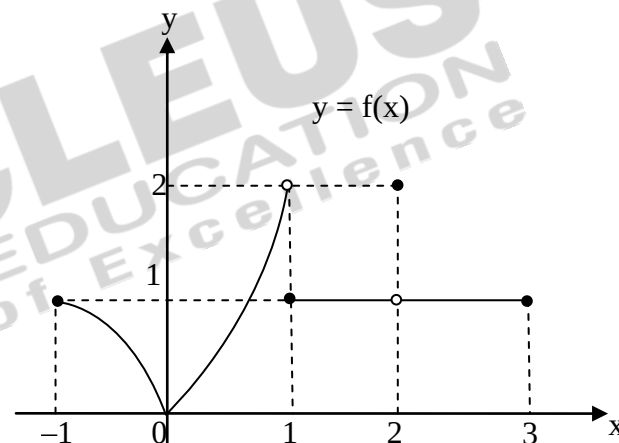
Solution :

- (a) As $x \rightarrow 0^-$: limit does not exist (the function is not defined to the left of $x = 0$)
- (b) As $x \rightarrow 0^+$: $f(x) \rightarrow -1 \Rightarrow \lim_{x \rightarrow 0^+} f(x) = -1$.
- (c) As $x \rightarrow 1^-$: $f(x) \rightarrow 1 \Rightarrow \lim_{x \rightarrow 1^-} f(x) = 1$.
- (d) As $x \rightarrow 1^+$: $f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 1^+} f(x) = 2$.
- (e) As $x \rightarrow 2^-$: $f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 2^-} f(x) = 3$.
- (f) As $x \rightarrow 2^+$: $f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 2^+} f(x) = 3$.

- (g) As $x \rightarrow 3^-$: $f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 3^-} f(x) = 2$.
- (h) As $x \rightarrow 3^+$: $f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 3^+} f(x) = 3$.
- (i) As $x \rightarrow 4^-$: $f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^-} f(x) = 4$.
- (j) As $x \rightarrow 4^+$: $f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^+} f(x) = 4$.
- (k) As $x \rightarrow \infty$: $f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow \infty} f(x) = 2$.
- (l) As $x \rightarrow 6^-$, $f(x) \rightarrow -\infty$
 $\Rightarrow \lim_{x \rightarrow 6^-} f(x) = -\infty$ limit does not exist because it is not finite.

Do yourself – 1 :

- (i) Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false ?



- (a) $\lim_{x \rightarrow -1^+} f(x) = 1$
- (b) $\lim_{x \rightarrow 2} f(x)$ does not exist
- (c) $\lim_{x \rightarrow 2} f(x) = 2$
- (d) $\lim_{x \rightarrow 1^-} f(x) = 2$
- (e) $\lim_{x \rightarrow 1} f(x)$ does not exist
- (f) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
- (g) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (-1, 1)$
- (h) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (1, 3)$
- (i) $\lim_{x \rightarrow 1^-} f(x) = 0$
- (j) $\lim_{x \rightarrow 3^+} f(x)$ does not exist.

4. FUNDAMENTAL THEOREMS ON LIMITS :

Let $\lim_{x \rightarrow a} f(x) = \ell$ and $\lim_{x \rightarrow a} g(x) = m$. If

ℓ and m exist finitely then :

- (a) Sum rule : $\lim_{x \rightarrow a} \{f(x) + g(x)\} = \ell + m$
- (b) Difference rule :
 $\lim_{x \rightarrow a} \{f(x) - g(x)\} = \ell - m$
- (c) Product rule : $\lim_{x \rightarrow a} f(x) \cdot g(x) = \ell \cdot m$
- (d) Quotient rule : $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\ell}{m}$,
provided $m \neq 0$
- (e) Constant multiple rule : $\lim_{x \rightarrow a} k f(x) = k$
 $\lim_{x \rightarrow a} f(x)$; where k is constant.
- (f) Power rule : If m and n are integers then
 $\lim_{x \rightarrow a} [f(x)]^{m/n} = \ell^{m/n}$ provided $\ell^{m/n}$ is a
real number.
- (g) $\lim_{x \rightarrow a} f[g(x)] = f\left(\lim_{x \rightarrow a} g(x)\right) = f(m)$;
provided $f(x)$ is continuous at $x = m$.
For example : $\lim_{x \rightarrow a} \ell^n g(x) = \ell^n \left[\lim_{x \rightarrow a} g(x) \right]$
 $= \ell^n (m)$; provided $\ell^n x$ is
continuous at $x = m$, $m = \lim_{x \rightarrow a} g(x)$.

5. INDETERMINATE FORMS :

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^0, \infty^0$$

Initially we will deal with first five forms only and the other two forms will come up after we have gone through differentiation.

Note :

- (i) Here 0,1 are not exact, infact both are approaching to their corresponding values.
- (ii) We cannot plot ∞ on the paper. Infinity (∞) is a symbol and not a number It does not obey the laws of elementary algebra,

$$\begin{aligned} (a) \infty + \infty &\rightarrow \infty & (b) \infty \times \infty &\rightarrow \infty \\ (c) \infty^\infty &\rightarrow \infty & (d) 0^\infty &\rightarrow 0 \end{aligned}$$

6. GENERAL METHODS TO BE USED TO EVALUATE LIMITS :

(A) Factorization :

Important factors :

- (i) $x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + \dots + a^{n-1})$, $n \in \mathbb{N}$
- (ii) $x^n + a^n = (x + a)(x^{n-1} - ax^{n-2} + \dots - a^{n-1})$, n is an odd natural number.

Note : $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$

Illustration 2:

Evaluate : $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right]$

Solution :

We have

$$\begin{aligned} &\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x(x-1) - 2(2x-3)}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x^2 - 5x + 6}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{(x-2)(x-3)}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x-3}{x(x-1)} \right] = -\frac{1}{2} \end{aligned}$$

Do yourself – 2 :

(i) Evaluate : $\lim_{x \rightarrow 1} \frac{x-1}{2x^2 - 7x + 5}$

(b) Rationalization or double rationalization :

Illustration 3:

Evaluate : $\lim_{x \rightarrow 1} \frac{4 - \sqrt{15x+1}}{2 - \sqrt{3x+1}}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 1} \frac{4 - \sqrt{15x+1}}{2 - \sqrt{3x+1}} \\ &= \lim_{x \rightarrow 1} \frac{(4 - \sqrt{15x+1})(2 + \sqrt{3x+1})(4 + \sqrt{15x+1})}{(2 - \sqrt{3x+1})(4 + \sqrt{15x+1})(2 + \sqrt{3x+1})} \\ &= \lim_{x \rightarrow 1} \frac{(15 - 15x)}{(3 - 3x)} \times \frac{2 + \sqrt{3x+1}}{4 + \sqrt{15x+1}} = \frac{5}{2} \end{aligned}$$

Illustration 4:

Evaluate : $\lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+8} - \sqrt{10-x^2}}{\sqrt{x^2+3} - \sqrt{5-x^2}} \right)$

Solution :

This is of the form $\frac{3-3}{2-2} = \frac{0}{0}$ if we put $x = 1$

To eliminate the $\frac{0}{0}$ factor, multiply by the conjugate of numerator and the conjugate of the denominator

$$\begin{aligned} \therefore \text{Limit} &= \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+8} - \sqrt{10-x^2}}{\sqrt{x^2+3} - \sqrt{5-x^2}} \right) \\ &= \frac{(\sqrt{x^2+8} + \sqrt{10-x^2})}{(\sqrt{x^2+3} + \sqrt{5-x^2})} \\ &\times \frac{(\sqrt{x^2+3} + \sqrt{5-x^2})}{(\sqrt{x^2+3} - \sqrt{5-x^2})} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3} + \sqrt{5-x^2}}{\sqrt{x^2+8} + \sqrt{10-x^2}} \\ &\times \frac{(x^2+8) - (10-x^2)}{(x^2+3) - (5-x^2)} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+3} + \sqrt{5-x^2}}{\sqrt{x^2+8} + \sqrt{10-x^2}} \right) \times 1 \\ &= \frac{2+2}{3+3} = \frac{2}{3} \end{aligned}$$

Do yourself - 3 :

(i) Evaluate : $\lim_{x \rightarrow 0} \frac{\sqrt{p+x} - \sqrt{p-x}}{\sqrt{q+x} - \sqrt{q-x}}$

(ii) Evaluate : $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$, $a \neq 0$

(iii) If $G(x) = -\sqrt{25-x^2}$, then find the $\lim_{x \rightarrow 1} \left(\frac{G(x) - G(1)}{x - 1} \right)$

(c) **Limit when $x \rightarrow \infty$:**

(i) Divide by greatest power of x in numerator and denominator.

(ii) Put $x = 1/y$ and apply $y \rightarrow 0$

Illustration 5 :

Evaluate : $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5}$

Solution :

$$\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Put $x = \frac{1}{y}$

$$\text{Limit} = \lim_{y \rightarrow 0} \frac{1 + y + y^2}{3 + 2y - 5y^2} = \frac{1}{3}$$

Illustration 6:

If $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$, then

(A) $a = 1, b = 1$

(B) $a = 1, b = 2$

(C) $a = 1, b = -2$

(D) none of these

Solution : $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$

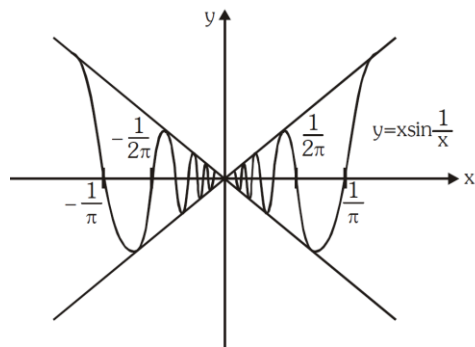
$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^3(1-a) - bx^2 - ax + (1-b)}{x^2 + 1} = 2$$

$$\lim_{x \rightarrow \infty} \frac{x(1-a) - b - \frac{a}{x} + \frac{(1-b)}{x^2}}{1 + \frac{1}{x^2}} = 2$$

$$\Rightarrow 1 - a = 0, -b = 2 \Rightarrow a = 1, b = -2$$

Ans.(C)

Ex. 2



$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$\therefore \sin\left(\frac{1}{x}\right)$ lies between -1 and 1

$$\Rightarrow -x \leq x \sin \frac{1}{x} \leq x$$

$$\Rightarrow \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 \text{ as } \lim_{x \rightarrow 0} (-x) = \lim_{x \rightarrow 0} x = 0$$

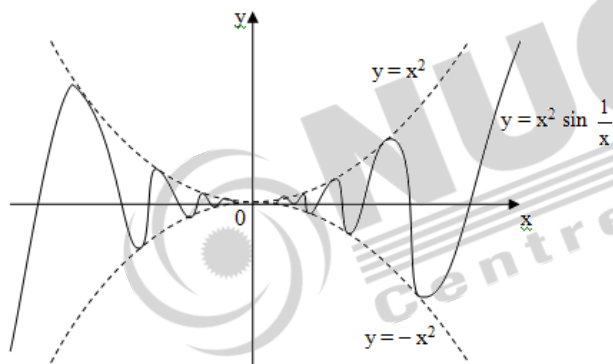
Do yourself -4 :

(i) Evaluate : $\lim_{n \rightarrow \infty} \frac{n+2 + \sqrt{n+1}}{n+2 - \sqrt{n+1}}$

(ii) Evaluate : $\lim_{n \rightarrow \infty} (n - \sqrt{n^2 + n})$

(d) **Squeeze play theorem (Sandwich theorem) :**

Statement : If $f(x) \leq g(x) \leq h(x)$; $\forall x$ in the neighbourhood at $x = a$ and



$$\lim_{x \rightarrow a} f(x) = \ell = \lim_{x \rightarrow a} h(x) \text{ then}$$

$$\lim_{x \rightarrow a} g(x) = \ell,$$

Ex. 1 $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0,$

$\therefore \sin\left(\frac{1}{x}\right)$ lies between -1 and 1

$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

$$\Rightarrow \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

$$\text{as } \lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$$

Illustration 7:

$$\text{Evaluate: } \lim_{n \rightarrow \infty} \frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2}$$

(where $[.]$ denotes the greatest integer function.)

Solution :

We know that $x - 1 < [x] \leq x$

$$\Rightarrow x + 2x + \dots + nx$$

$$-n < \sum_{r=1}^n [rx] \leq x + 2x + \dots + nx$$

$$\Rightarrow \frac{xn}{2}(n+1) - n < \sum_{r=1}^n [rx] \leq \frac{x.n(n+1)}{2}$$

$$\Rightarrow \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} < \frac{1}{n^2} \sum_{r=1}^n [rx] \leq \frac{x}{2} \left(1 + \frac{1}{n}\right)$$

$$\text{Now, } \lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) = \frac{x}{2} \text{ and}$$

$$\lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} = \frac{x}{2}$$

$$\text{Thus, } \lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2} = \frac{x}{2}$$

7. LIMIT OF TRIGONOMETRIC FUNCTIONS :

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} \quad [\text{where } x \text{ is measured in radians}]$$

(a) If $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} \frac{\sin f(x)}{f(x)} = 1$,

e.g. $\lim_{x \rightarrow 1} \frac{\sin(\ell nx)}{(\ell nx)} = 1$

Illustration 8:

Evaluate : $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x^3 \cos x}{\sin x (1 - \cos x)} \\ &= \lim_{x \rightarrow 0} \frac{x^3 \cos x (1 + \cos x)}{\sin x \cdot \sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{x^3}{\sin^3 x} \cdot \cos x (1 + \cos x) = 2 \end{aligned}$$

Illustration 9:

Evaluate : $\lim_{x \rightarrow 0} \frac{(2+x) \sin(2+x) - 2 \sin 2}{x}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{2(\sin(2+x) - \sin 2) + x \sin(2+x)}{x} \\ &= \lim_{x \rightarrow 0} \left(\frac{2 \cdot 2 \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{x} + \sin(2+x) \right) \\ &= \lim_{x \rightarrow 0} \frac{2 \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{\frac{x}{2}} \\ &+ \lim_{x \rightarrow 0} \sin(2+x) = 2 \cos 2 + \sin 2 \end{aligned}$$

Illustration 10:

Evaluate : $\lim_{n \rightarrow \infty} \frac{\sin \frac{a}{n}}{\tan \frac{b}{n+1}}$

Solution :

As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ and $\frac{a}{n}$ also tends to zero

$\sin \frac{a}{n}$ should be written as $\frac{\sin \frac{a}{n}}{\frac{a}{n}}$

so that it looks like $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$

The given limit

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a(n+1)}{n \cdot b} \\ &= \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a}{b} \left(1 + \frac{1}{n} \right) \\ &= 1 \times 1 \times \frac{a}{b} \times 1 = \frac{a}{b} \end{aligned}$$

Do yourself – 5:

(i) Evaluate :

(a) $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\tan \beta x}$

(b) $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$

(c) $\lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$

8. LIMIT OF EXPONENTIAL FUNCTIONS :

(a) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ell n a$ ($a > 0$) In particular

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

In general if $\lim_{x \rightarrow a} f(x) = 0$, then

$$\lim_{x \rightarrow a} \frac{a^{f(x)} - 1}{f(x)} = \ln a, a > 0$$

Illustration 11:

Evaluate : $\lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} \\ &= \lim_{x \rightarrow 0} \frac{e^x \times e^{(\tan x - x)} - e^x}{\tan x - x} \\ &= \lim_{x \rightarrow 0} \frac{e^x (e^{\tan x - x} - 1)}{\tan x - x} = \lim_{x \rightarrow 0} \frac{e^x (e^y - 1)}{y} \\ & \text{where } y = \tan x - x \text{ and } \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \\ &= e^0 \times 1 \\ &[\text{as } x \rightarrow 0, \tan x - x \rightarrow 0] \\ &= 1 \times 1 = 1 \end{aligned}$$

Do yourself – 6 :

(i) Evaluate : $\lim_{x \rightarrow a} \frac{e^x - e^a}{x - a}$

(ii) Evaluate : $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$

(b) (i) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

(Note : The base and exponent depends on the same variable.) In general, if

$\lim_{x \rightarrow a} f(x) = 0$, then

$\lim_{x \rightarrow a} (1 + f(x))^{1/f(x)} = e$

(ii) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

(iii) If $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} \phi(x) = \infty$, then

$\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^k$

where $k = \lim_{x \rightarrow a} \phi(x) [f(x) - 1]$

Illustration 12:

Evaluate $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3} \\ &= \lim_{x \rightarrow 1} (\log_3 3 + \log_3 x)^{\log_x 3} \\ &= \lim_{x \rightarrow 1} (1 + \log_3 x)^{1/\log_3 x} = e \\ &\therefore \log_b a = \frac{1}{\log_a b} \end{aligned}$$

Illustration 13:

Evaluate : $\lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{1 - \cos x}$

Solution :

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{1 - \cos x} \\ &= \lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{\frac{1 - \cos x}{x^2} \cdot x^2} \cdot \frac{2 \tan x}{2 \tan x} = 4 \end{aligned}$$

Illustration 14 :

Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2}$

Solution :

Since it is in the form of 1^∞

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2} \\ &= e^{\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1 - 2x^2 - 3}{2x^2 + 3} \right) (4x^2 + 2)} = e^{-8} \end{aligned}$$

Do yourself – 7 :

(i) Evaluate : $\lim_{x \rightarrow \infty} x \{ \ln(x+a) - \ln x \}$

(ii) Evaluate : $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{pn+q}$

(iii) Evaluate : $\lim_{x \rightarrow 0} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}}$

(iv) Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$

- (c) If $\lim_{x \rightarrow a} f(x) = A > 0$ and $\lim_{x \rightarrow a} \phi(x) = B$,
then $\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^{B \ln A} = A^B$

Illustration 15:

Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{7x^2 + 1}{5x^2 - 1} \right)^{\frac{x^5}{1-x^3}}$

Solution :

Here $f(x) = \frac{7x^2 + 1}{5x^2 - 1}$,

$\phi(x) = \frac{x^5}{1-x^3} = \frac{x^2 \cdot x^3}{1-x^3} = \frac{x^2}{\frac{1}{x^3} - 1}$

$\therefore \lim_{x \rightarrow \infty} f(x) = \frac{7}{5}$ and $\lim_{x \rightarrow \infty} \phi(x) \rightarrow \infty$

$\Rightarrow \lim_{x \rightarrow \infty} (f(x))^{\phi(x)} = \left(\frac{7}{5} \right)^{-\infty} = 0$

Do yourself – 8 :

(i) Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{1+5x^2}{1+3x^2} \right)^{-x^2}$

9. LIMIT USING SERIES EXPANSION :

Expansion of function like binomial expansion, exponential and logarithmic expansion, expansion of $\sin x$, $\cos x$, $\tan x$ should be remembered by heart which are given below :

- (a) $a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \dots$
 $a > 0$
- (b) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
- (c) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ for $-1 < x \leq 1$
- (d) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

(e) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

(f) $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

(g) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$

(h) $\sin^{-1} x$
 $= x + \frac{1^2}{3!} x^3 + \frac{1^2 \cdot 3^2}{5!} x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!} x^7 + \dots$

(i) $\sec^{-1} x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \dots$

(j) $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$
 $\dots \dots \dots n \in \mathbb{Q}$

Illustration 16:

$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

Solution :

$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$
 $\Rightarrow \lim_{x \rightarrow 0} \frac{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\dots - 1-x+\frac{x^3}{2!}-\frac{x^3}{3!}+\dots - 2x}{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)}$

$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \cdot \frac{x^3}{6} + 2 \cdot \frac{x^5}{5!} + \dots}{\frac{x^3}{6} + \frac{x^5}{5!} - \dots}$

$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{1}{3} + \frac{1}{60} x^2 + \dots \right)}{x^3 \left(\frac{1}{6} + \frac{1}{120} x^2 + \dots \right)} = \frac{1/3}{1/6} = 2$

Do yourself – 9 :

(i) Evaluate : $\lim_{x \rightarrow 0} \frac{x - \sin x}{\sin(x^3)}$

(ii) Evaluate : $\lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}$

Miscellaneous Illustrations :

Illustration 17 :

Evaluate $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

Solution :

Again the function $f(x) = \sin (\pi/x)$ is undefined at 0. Evaluating the function for some small values of x , we get

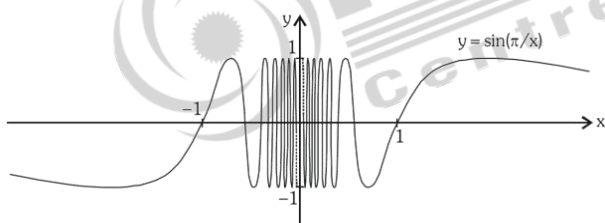
$$f(1) = \sin \pi = 0, f\left(\frac{1}{2}\right) = \sin 2\pi = 0,$$

$$f(0.1) = \sin 10 \pi = 0, f(0.01) = \sin 100\pi = 0.$$

On the basis of this information we might be tempted to guess that $\lim_{x \rightarrow 0} \sin$

$$\frac{\pi}{x} = 0 \text{ but this time our guess is wrong.}$$

Note that although $f(1/n) = \sin n\pi = 0$ for any integer n , it is also true that $f(x) = 1$ for infinitely many values of x that approach 0. [In fact, $\sin (\pi/x) = 1$ when $\frac{\pi}{x} = \frac{\pi}{2} + 2n\pi$ and solving for x , we get $x = 2/(4n + 1)$]. The graph of f is given in following figure



The dashed line indicate that the values of $\sin (\pi/x)$ oscillate between 1 and -1 infinitely often as x approaches 0. Since the values of $f(x)$ do not approach a fixed number as x approaches 0,

$$\Rightarrow \lim_{x \rightarrow 0} \sin \frac{\pi}{x} \text{ does not exist.}$$

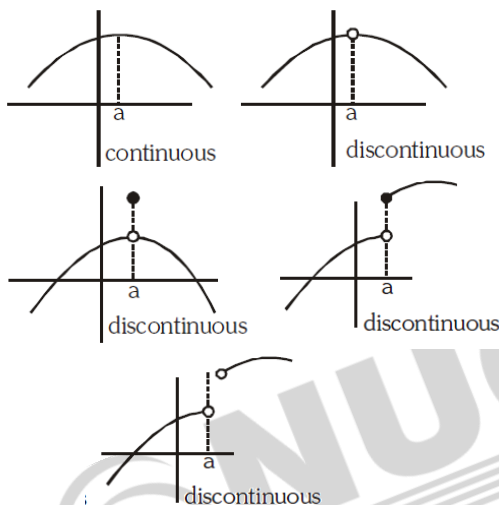
ANSWERS FOR DO YOURSELF

- | | | | | | |
|-----|-------|------------------------------|-------|-----------------------|---------------|
| 1 : | (i) | (a) T | (b) F | (c) F | (d) T |
| | | (e) T | (f) T | (g) T | (h) T |
| | | (i) F | (j) T | | |
| 2. | (i) | $-\frac{1}{3}$ | | | |
| 3: | (i) | $\frac{\sqrt{q}}{\sqrt{p}}$ | (ii) | $\frac{2}{3\sqrt{3}}$ | |
| | (iii) | $\frac{1}{\sqrt{24}}$ | | | |
| 4: | (i) | 1 | (ii) | $-\frac{1}{2}$ | |
| 5 : | (i) | (a) $\frac{\alpha}{\beta}$ | (b) | $\frac{\sin 2y}{2y}$ | |
| | | (c) $2a \sin a + a^2 \cos a$ | | | |
| 6. | (i) | e^a | (ii) | $2 \ln 2$ | |
| 7. | (i) | a | (ii) | e^p | |
| | (iii) | $e^{\frac{1}{2}}$ | (iv) | e^5 | |
| 8. | (i) | 0 | 9. | (i) | $\frac{1}{6}$ |
| | (ii) | $\frac{1}{3}$ | | | |

CONTINUITY

1. CONTINUOUS FUNCTIONS:

A function for which a small change in the independent variable causes only a small change and not a sudden jump in the dependent variable are called continuous functions. Naively, we may that a function is continuous at fixed point if we can draw the graph of the function around the point without lifting the pen from the plane of the paper.



Continuity of a function at a point :

A function $f(x)$ is said to be continuous at $x = a$, if $\lim_{x \rightarrow a} f(x) = f(a)$.

Symbolically f is continuous at $x = a$ if $\lim_{h \rightarrow 0} f(a - h) = \lim_{h \rightarrow 0} f(a + h) = f(a)$, $h > 0$

i. e. $(LHL)_{x=a} = (RHL)$ equals value of ' f ' at $x = a$.

Illustration 1:

If $f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x] & x \geq 1 \end{cases}$ then find whether $f(x)$ is continuous or not at $x = 1$, where $[]$ denotes greatest integer function.

Solution :

$$f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x] & x \geq 1 \end{cases}$$

For continuity at $x = 1$, we determine, $f(1)$, $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.

Now, $f(1) = [1] = 1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \sin \frac{\pi x}{2} = \sin \frac{\pi}{2} = 1$$

$$\text{and } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} [x] = 1$$

$$\text{So } f(1) = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$\therefore f(x)$ is continuous at $x = 1$

Illustration 2:

Let

$$f(x) = \begin{cases} \frac{a(1 - x \sin x) + b \cos x + 5}{x^2} & x < 0 \\ 3 & x = 0 \\ \left(01 + \left(\frac{cx + dx^3}{x^2} \right) \right)^{\frac{1}{x}} & x > 0 \end{cases}$$

If f is continuous at $x = 0$, then find out the values of a , b , c and d .

Solution :

Since $f(x)$ is continuous at $x = 0$, so at $x = 0$, both left and right limits must exist and both must be equal to 3.

$$\text{Now } \lim_{x \rightarrow 0^-} \frac{a(1 - x \sin x) + b \cos x + 5}{x^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{(a + b + 5) + \left(-a - \frac{b}{2} \right) x^2 + \dots}{x^2} = 3$$

(By the expansion of $\sin x$ and $\cos x$)

If $\lim_{x \rightarrow 0^-} f(x)$ exists then $a + b + 5 = 0$

$$\text{and } -a - \frac{b}{2} = 3 \Rightarrow a = -1 \text{ and } b = -4$$

$$\text{since } \lim_{x \rightarrow 0^+} \left(1 + \left(\frac{cx + dx^3}{x^2} \right) \right)^{\frac{1}{x}} \text{ exists}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{cx + dx^3}{x^2} = 0 \Rightarrow c = 0$$

$$\text{Now } \lim_{x \rightarrow 0^+} (1 + dx)^{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \left[(1 + dx)^{\frac{1}{dx}} \right]^d$$

$$= e^d$$

$$\text{So } e^d = 3 \Rightarrow d = \ln 3,$$

Hence $a = -1$, $b = -4$, $c = 0$ and $d = \ln 3$.

Do yourself - 1 :

(i) If $f(x) = \begin{cases} \cos x; & x \geq 0 \\ x + k; & x < 0 \end{cases}$ find the value of k if $f(x)$ is continuous at $x = 0$.

(ii) If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)} & ; x \neq -2 \\ 2 & ; x = -2 \end{cases}$ then discuss the continuity of $f(x)$ at $x = -2$

2. CONTINUITY OF THE FUNCTION IN AN INTERVAL :

(a) A function is said to be continuous in (a, b) if f is continuous at each and every point belonging to (a, b) .

(b) A function is said to be continuous in a closed interval $[a, b]$ if :

(i) f is continuous in the open interval (a, b)

(ii) f is right continuous at 'a'
i.e. $\lim_{x \rightarrow a^+} f(x) = f(a) = \text{a finite quantity}$

(iii) f is left continuous at 'b'
i.e. $\lim_{x \rightarrow b^-} f(x) = f(b) = \text{a finite quantity}$

Note :

(i) All polynomials, trigonometrical functions, exponential and logarithmic functions are continuous in their domains.

(ii) If $f(x)$ and $g(x)$ are two functions that are continuous at $x = c$ then the function defined by :

$F_1(x) = f(x) \pm g(x)$; $F_2(x) = Kf(x)$,
where K is any real number ;
 $F_3(x) = f(x) \cdot g(x)$ are also continuous at $x = c$.

Further, if $g(c)$ is not zero, then $F_4(x) = \frac{f(x)}{g(x)}$ is also continuous at $x = c$.

Illustration 3:

Discuss the continuity of

$$f(x) = \begin{cases} |x+1| & , x < -2 \\ 2x+3 & , -2 \leq x < 0 \\ x^2+3 & , 0 \leq x < 3 \\ x^3-15 & , x \geq 3 \end{cases}$$

Solution :

We write $f(x)$ as

$$f(x) = \begin{cases} -x-1 & , x < -2 \\ 2x+3 & , -2 \leq x < 0 \\ x^2+3 & , 0 \leq x < 3 \\ x^3-15 & , x \geq 3 \end{cases}$$

As we can see, $f(x)$ is defined as a polynomial function in each of intervals $(-\infty, -2)$, $(-2, 0)$, $(0, 3)$ and $(3, \infty)$. Therefore, it is continuous in each of these four open intervals. Thus we check the continuity at $x = -2, 0, 3$.

At the point $x = -2$

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (-x-1) = +2-1 = 1$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (2x+3) = 2 \cdot (-2) + 3 = -1$$

Therefore, $\lim_{x \rightarrow -2} f(x)$ does not exist and

hence $f(x)$ is discontinuous at $x = -2$.

At the point $x = 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (2x+3) = 3$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2+3) = 3$$

$$f(0) = 0^2 + 3 = 3$$

Therefore $f(x)$ is continuous at $x = 0$.

At the point $x = 3$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2+3) = 3^2 + 3 = 12$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x^3-15) = 3^3 - 15 = 12$$

$$f(3) = 3^3 - 15 = 12$$

Therefore, $f(x)$ is continuous at $x = 3$.

We find that $f(x)$ is continuous at all points in $\mathbb{R}$ except at $x = -2$

Do yourself - 2:

(i) If $f(x) = \begin{cases} \frac{x^2}{a} & ; 0 \leq x < 1 \\ -1 & ; 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^2} & ; \sqrt{2} \leq x < \infty \end{cases}$ then

find the value of a and b if $f(x)$ is continuous in $[0, \infty)$

(ii) Discuss the continuity of

$$f(x) = \begin{cases} |x-3| & ; 0 \leq x < 1 \\ \sin x & ; 1 \leq x \leq \frac{\pi}{2} \\ \log_{\frac{\pi}{2}} x & ; \frac{\pi}{2} < x < 3 \end{cases} \text{ in } [0, 3]$$

3. TYPES OF DISCONTINUITIES :

Type-1 : (Removable type of discontinuities)

: - In case $\lim_{x \rightarrow a} f(x)$ exists but is not equal to $f(a)$ then the function is said to have a removable discontinuity or discontinuity of the first kind. In this case we can redefine the function such that $\lim_{x \rightarrow a} f(x) = f(a)$ and make it continuous at $x = a$. **Removable type of discontinuity can be further classified as :**

(a) **Missing point discontinuity :**

Where $\lim_{x \rightarrow a} f(x)$ exists but $f(a)$ is not defined.

(b) **Isolated point discontinuity :**

Where $\lim_{x \rightarrow a} f(x)$ exists and $f(a)$ also exists but ; $\lim_{x \rightarrow a} f(x) \neq f(a)$.

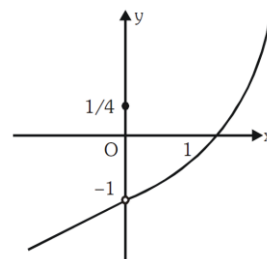
Illustration 4:

Examine the function,

$$f(x) = \begin{cases} x-1 & , x < 0 \\ 1/4 & , x = 0 \\ x^2 - 1 & , x > 0 \end{cases}$$

Discuss the continuity, and if discontinuous remove the discontinuity by redefining the function (if possible).

Solution :



$y = f(x)$ before redefining

Graph of $f(x)$ is shown, from graph it is seen that

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = -1,$$

$$\text{but } f(0) = 1/4$$

Thus, $f(x)$ has removable discontinuity and $f(x)$ could be made continuous by taking $f(0) = -1$

$$\Rightarrow f(x) = \begin{cases} x-1 & , x < 0 \\ -1 & , x = 0 \\ x^2 - 1 & , x > 0 \end{cases}$$

Do yourself - 3:

(i) If $f(x) = \begin{cases} \frac{1}{x-1} & ; 0 \leq x < 2 \\ x^2 - 3 & ; 2 \leq x < 4 \\ 5 & ; x = 4 \\ 14 - \frac{x^{1/2}}{2} & ; x > 4 \end{cases}$, then

discuss the types of discontinuity for the function.

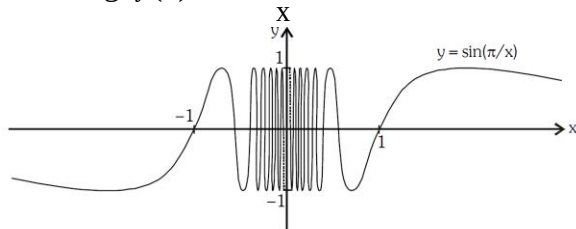
Type - 2 :

(Non-Removable type of discontinuities) :

In case $\lim_{x \rightarrow a} f(x)$ does not exist then it is not possible to make the function continuous by redefining it. Such a discontinuity is known as non-removable discontinuity or discontinuity of the 2nd kind. Non-removable type of discontinuity can be further classified as :

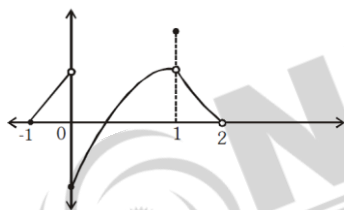
- (i) **Finite type discontinuity** : In such type of discontinuity left hand limit and right hand limit at a point exists but are not equal.
- (ii) **Infinite type discontinuity** : In such type of discontinuity at least one of the limit viz. LHL and RHL is tending to infinity.
- (iii) **Oscillatory type discontinuity** :

e.g. $f(x) = \sin \frac{\pi}{x}$ at $x = 0$



$f(x)$ has non removable oscillatory type discontinuity at $x = 0$

Example :



From the adjacent graph note that

- f is continuous at $x = -1$
- f has isolated discontinuity at $x = 1$
- f has missing point discontinuity at $x = 2$
- f has non removable (finite type) discontinuity at the origin.

Note : In case of non-removable (finite type) discontinuity the non-negative difference between the value of the RHL at $x = a$ and LHL at $x = a$ is called the jump of discontinuity. A function having a finite number of jumps in a given interval I is called a **piece wise continuous or sectionally continuous function** in this interval.

Illustration 5:

Show that the function,

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & ; \text{ when } x \neq 0 \\ 0, & ; \text{ when } x = 0 \end{cases}$$

has non-removable discontinuity at $x = 0$.

Solution :

We have,

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & ; \text{ when } x \neq 0 \\ 0, & ; \text{ when } x = 0 \end{cases}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{1 - \frac{1}{e^{1/h}}}{1 + \frac{1}{e^{1/h}}} = 1 \quad [\because e^{1/h} \rightarrow \infty]$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1} = \frac{0 - 1}{0 + 1} = -1$$

[$\because h \rightarrow 0; e^{-1/h} \rightarrow 0$]

$$\lim_{x \rightarrow 0^-} f(x) = -1$$

$\Rightarrow \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$. Thus $f(x)$ has non-removable discontinuity.

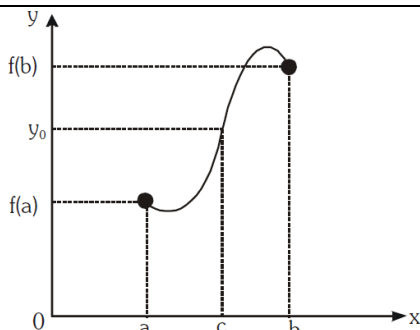
Do yourself - 4:

- (i) Discuss the type of discontinuity for

$$f(x) = \begin{cases} -1 & ; x \leq -1 \\ |x| & ; -1 < x < 1 \\ (x+1) & ; x \geq 1 \end{cases}$$

4. THE INTERMEDIATE VALUE THEOREM :

Suppose $f(x)$ is continuous on an interval I , and a and b are any two points of I . Then if y_0 is a number between $f(a)$ and $f(b)$, there exists a number c between a and b such that $f(c) = y_0$



The function f , being continuous on $[a, b]$ takes on every value between $f(a)$ and $f(b)$

Note that a function f which is continuous in $[a, b]$ possesses the following properties :

- (i) If $f(a)$ and $f(b)$ possess opposite signs, then there exists at least one root of the equation $f(x) = 0$ in the open interval (a, b) .
- (ii) If K is any real number between $f(a)$ and $f(b)$, then there exists at least one root of the equation $f(x) = K$ in the open interval (a, b) .

Note : In above cases the number of roots is always odd.

Illustration 6 :

Show that the function, $f(x) = (x-a)^2(x-b)^2 + x$, takes the value $\frac{a+b}{2}$ for some $x_0 \in (a, b)$

Solution :

$$f(x) = (x-a)^2(x-b)^2 + x$$

$$f(a) = a$$

$$f(b) = b \text{ and } \frac{a+b}{2} \in (f(a), f(b))$$

$\therefore$ By intermediate value theorem, there is at least one $x_0 \in (a, b)$ such that

$$f(x_0) = \frac{a+b}{2}.$$

Illustration 7:

Let $f : [0, 1] \xrightarrow{\text{onto}} [0, 1]$ be a continuous function, then prove that $f(x) = x$ for at least one $x \in [0, 1]$

Solution :

$$\text{Consider } g(x) = f(x) - x$$

$$g(0) = f(0) - 0 = f(0) \geq 0$$

$$\{ \because 0 \leq f(x) \leq 1 \}$$

$$g(1) = f(1) - 1 \leq 0$$

$$\Rightarrow g(0) \cdot g(1) \leq 0$$

$$\Rightarrow g(x) = 0 \text{ has at least one root in } [0, 1]$$

$$\Rightarrow f(x) = x \text{ for at least one } x \in [0, 1]$$

Do yourself - 5:

- (i) If $f(x)$ is continuous in $[a, b]$ such that $f(c) = \frac{2f(a)+3f(b)}{5}$, then prove that $c \in (a, b)$

5. SOME IMPORTANT POINTS :

- (a) If $f(x)$ continuous and $g(x)$ is discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ will not necessarily be discontinuous at $x = a$, e.g.

$$f(x) = x \text{ and } g(x) = \begin{cases} \sin \frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$ is continuous at $x = 0$ and $g(x)$ is discontinuous at $x = 0$. but $f(x) \cdot g(x)$ is continuous at $x = 0$.

- (b) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ is not necessarily be discontinuous at $x = a$, e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

$f(x)$ and $g(x)$ both are discontinuous at $x = 0$ but the product function $f(x) \cdot g(x)$ is still continuous at $x = 0$

- (c) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then $f(x) \pm g(x)$ is not necessarily be discontinuous at $x = a$

- (d) A continuous function whose domain is closed must have a range also in closed interval.

- (e) If f is continuous at $x = a$ and g is continuous at $x = f(a)$ then the composite $g[f(x)]$ is continuous at $x = a$. eg. $f(x) = \frac{x \sin x}{x^2 + 2}$ and $g(x) = |x|$ are continuous at $x = 0$, hence the composite $(g \circ f)(x) = \left| \frac{x \sin x}{x^2 + 2} \right|$ will also be continuous at $x = 0$

Illustration 8:

If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{1}{x-2}$, then discuss the continuity of $f(x)$, $g(x)$ and $f \circ g(x)$ in $\mathbb{R}$.

Solution :

$$f(x) = \frac{x+1}{x-1}$$

$f(x)$ is a rational function it must be continuous in its domain and f is not defined at $x = 1$.

$\therefore f$ is discontinuous at $x = 1$

$$g(x) = \frac{1}{x-2}$$

$g(x)$ is also a rational function. It must be continuous in its domain and g is not defined at $x = 2$.

$\therefore g$ is discontinuous at $x = 2$

Now $f \circ g(x)$ will be discontinuous at $x = 2$ (point of discontinuity of $g(x)$)

Consider $g(x) = 1$ (when $g(x)$ = point of discontinuity of $f(x)$)

$$\frac{1}{x-2} = 1 \Rightarrow x = 3$$

$\therefore f \circ g(x)$ is discontinuous at $x = 2$ and $x = 3$.

Do yourself - 6 :

- (i) Let $f(x) = [x]$ and $g(x) = \text{sgn}(x)$ (where $[.]$ denotes greatest integer function), then discuss the continuity of $f(x) \pm g(x)$, $f(x) \cdot g(x)$ and $\frac{f(x)}{g(x)}$ at $x = 0$.

- (ii) If $f(x) = \sin |x|$ and $g(x) = \tan |x|$ then discuss the continuity of $f(x) \pm g(x)$; $\frac{f(x)}{g(x)}$ and $f(x) \cdot g(x)$

6. SINGLE POINT CONTINUITY :

Functions which are continuous only at one point are said to exhibit single point continuity

Illustration 9:

If $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q} \end{cases}$, find the points where $f(x)$ is continuous

Solution :

Let $x = a$ be the point at which $f(x)$ is continuous.

$$\Rightarrow \lim_{\substack{x \rightarrow a \\ \text{through rational}}} f(x) = \lim_{\substack{x \rightarrow a \\ \text{through irrational}}} f(x)$$

$$\Rightarrow a = -a$$

$$\Rightarrow a = 0$$

$\Rightarrow$ function is continuous at $x = 0$.

Do yourself - 7:

- (i) If $g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.
- (ii) If $f(x) = \begin{cases} x^2 & ; x \in \mathbb{Q} \\ 1-x^2 & ; x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.

ANSWERS FOR DO YOURSELF

1. (i) 1
(ii) discontinuous at $x = -2$
2. (i) $a = -1$ and $b = 1$
(ii) Discontinuous at $x = 1$ and continuous at $x = \frac{\pi}{2}$
3. (i) Non-removable infinite type discontinuity at $x = 1$, isolated point removable discontinuity at $x = 4$.
4. (i) Finite type non-removable discontinuity at $x = -1, 1$
6. (i) All are discontinuous at $x = 0$.
(ii) $f(x)$, $g(x)$ and $f(x) \pm g(x)$ are discontinuous at $x = (2n + 1)\frac{\pi}{2}; n \in I$
 $\frac{f(x)}{g(x)}$ is discontinuous at $x = \frac{n\pi}{2}; n \in I$
7. $x = 0$ (ii) $x = \pm \frac{1}{\sqrt{2}}$

DIFFERENTIABILITY

1. MEANING OF DERIVATIVE :

The instantaneous rate of change of a function with respect to the dependent variable is called derivative. Let ' f ' be a given function of one variable and let Δx denote a number (positive or negative) to be added to the number x . Let Δf denote the corresponding change of ' f ' then

$$\Delta f = f(x + \Delta x) - f(x)$$

$$\Rightarrow \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

If $\Delta f/\Delta x$ approaches a limit as Δx approaches zero, this limit is the

derivative of ' f ' at the point x . The derivative of a function ' f ' is a function; this function is denoted by symbols such as

$$f'(x), \frac{df}{dx}, \frac{d}{dx} f(x) \text{ or } \frac{df(x)}{dx}$$

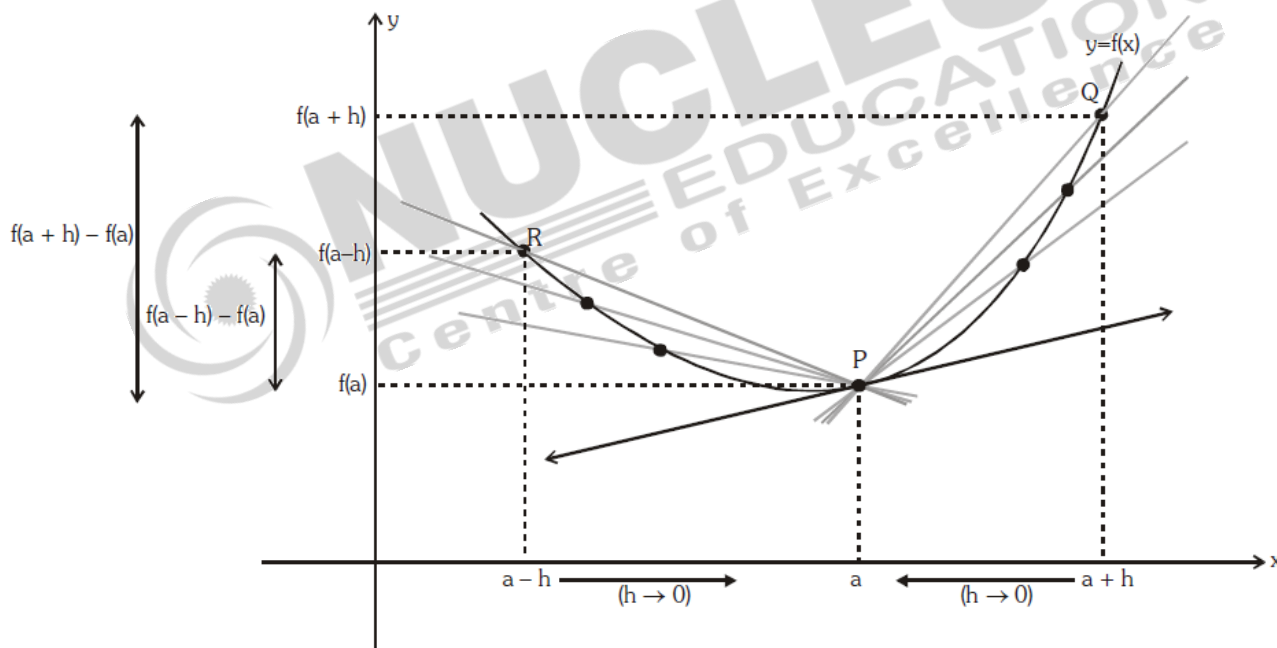
$$\Rightarrow \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The derivative evaluated at a point a , is

$$\text{written, } f'(a), \left. \frac{df(x)}{dx} \right|_{x=a}, f'(x)_{x=a}, \text{ etc.}$$

2. EXISTENCE OF DERIVATIVE AT $x = a$:



(a) Right hand derivative :

The right hand derivative of $f(x)$ at $x = a$ denoted by $f'(a^+)$ is defined as :

$$f'(a^+) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided}$$

the limit exists and is finite. ($h > 0$)

(b) Left hand derivative :

The left hand derivative of $f(x)$ at $x = a$ denoted by $f'(a^-)$ is defined as :

$$f'(a^-) = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}, \text{ provided}$$

the limit exists and is finite. ($h > 0$)

Hence $f(x)$ is said to be **derivable or differentiable** at $x = a$. If $f'(a^+) = f'(a^-) =$ finite quantity and it is denoted by $f'(a)$; where $f'(a) = f'(a^-) = f'(a^+)$ and it is called derivative or differential coefficient of $f(x)$ at $x = a$.

3. DIFFERENTIABILITY AND CONTINUITY :

Theorem : If a function $f(x)$ is derivable at $x = a$, then $f(x)$ is continuous at $x = a$.

Proof : $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

exists.

$$\text{Also } f(a+h) - f(a) = \frac{f(a+h) - f(a)}{h} \cdot h$$

$[h \neq 0]$

$$\therefore \lim_{h \rightarrow 0} [f(a+h) - f(a)]$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a) \cdot 0 = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} [f(a+h) - f(a)] = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} f(a+h) = f(a) \Rightarrow f(x) \text{ is continuous at } x = a.$$

Note :

- (i) $\boxed{\begin{array}{l} \text{Differentiable} \Rightarrow \text{Continuous ;} \\ \text{Continuity} \Rightarrow \text{Differentiable ; Not} \\ \text{Differentiable} \\ \Rightarrow \text{Not Continuous But Not} \\ \text{Continuous} \Rightarrow \text{Not Differentiable} \end{array}}$

(ii) All polynomial, trigonometric, logarithmic and exponential function are continuous and differentiable in their domains.

(iii) If $f(x)$ and $g(x)$ are differentiable at $x = a$ then the function $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \cdot g(x)$ will also be differentiable at $x = a$ and if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be differentiable at $x = a$.

Illustration 1 :

$$\text{Let } f(x) = \begin{cases} \operatorname{sgn}(x) + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$$

Discuss the continuity and differentiability at $x = 0$ and $\frac{\pi}{2}$.

Solution :

$$f(x) = \begin{cases} -1 + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$$

To check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-1+0-h - (-1)}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-1 + \sin h + 1}{h} = 1$$

$$\therefore \text{LHD} = \text{RHD}$$

$\therefore$ Differentiable at $x = 0$.

$\Rightarrow$ Continuous at $x = 0$.

To check the continuity at $x = \frac{\pi}{2}$

$$\text{LHL } \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} (-1 + \sin x) = 0$$

$$\text{RHL } \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} \cos x = 0$$

$$\therefore \text{LHL} = \text{RHL} = f\left(\frac{\pi}{2}\right) = 0$$

$\therefore$ Continuous at $x = \frac{\pi}{2}$.

To check the differentiability at $x = \frac{\pi}{2}$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-1 + \cos h - 0}{-h} = 0$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-\sinh - 0}{h} = -1$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ not differentiable at $x = \frac{\pi}{2}$.

Illustration 2:

$$\text{If } f(x) = \begin{cases} A + Bx^2 & x < 1 \\ 3Ax - B + 2x \geq 1 \end{cases}$$

then find A and B so that $f(x)$ become differentiable at $x = 1$.

Solution :

$$\begin{aligned} f'(1^+) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3A(1+h) - B + 2 - 3A + B - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{3Ah}{h} = 3A \end{aligned}$$

$$\begin{aligned} f'(1^-) &= \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{A + B(1-h^2) - 3A + B - 2}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(-2A + 2B - 2) + Bh^2 - 2Bh}{-h} \end{aligned}$$

hence for this limit to be defined

$$-2A + 2B - 2 = 0$$

$$B = A + 1$$

$$f'(1^-) = \lim_{h \rightarrow 0} -(Bh - 2B) = 2B$$

$$\therefore f'(1^-) = f'(1^+)$$

$$3A = 2B = 2(A + 1)$$

$$A = 2, B = 3$$

Illustration 3:

$$f(x) = \begin{cases} [\cos \pi x] & x \leq 1 \\ 2\{x\} - 1 & x > 1 \end{cases} \quad \text{comment on}$$

the derivability at $x = 1$, where $[]$ denotes greatest integer function and $\{ \}$ denotes fractional part function.

Solution :

$$f'(1^-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{[\cos(\pi - \pi h)] + 1}{-h} = \lim_{h \rightarrow 0} \frac{-1 + 1}{-h} = 0$$

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2\{1+h\} - 1 + 1}{h} = \lim_{h \rightarrow 0} \frac{2h}{h} = 2$$

Hence $f(x)$ is not differentiable at $x = 1$.

Do yourself - 1:

(i) A function is defined as follows :

$$f(x) = \begin{cases} x^3 & ; \quad x^2 < 1 \\ x & ; \quad x^2 \geq 1 \end{cases} \quad \text{discuss the}$$

continuity and differentiability at $x = 1$.

(ii)

If $f(x) = \begin{cases} ax^3 + b, & \text{for } 0 \leq x \leq 1 \\ 2\cos \pi x + \tan^{-1} x, & \text{for } 1 < x \leq 2 \end{cases}$ be the differentiable function in $[0, 2]$, then find a and b. (where $[.]$ denotes the greatest integer function)

4. IMPORTANT NOTE :

(a) Let $f'(a^+) = p$ and $f'(a^-) = q$ where p and q are finite then :

(i) $p = q \Rightarrow f$ is differentiable at $x = a$
 $\Rightarrow f$ is continuous at $x = a$

(ii) $p \neq q \Rightarrow f$ is not differentiable at $x = a$, but f is continuous at $x = a$.

Illustration 4 :

Determine the values of x for which the following functions fails to be continuous or differentiable

$$f(x) = \begin{cases} (1-x), & x < 1 \\ (1-x)(2-x), & 1 \leq x \leq 2 \\ (3-x), & x > 2 \end{cases}$$

Justify your answer.

Solution :

By the given definition it is clear that the function f is continuous and differentiable at all points except possibly at $x = 1$ and $x = 2$.

Check the differentiability at $x = 1$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{1 - (1-h) - 0}{-h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\{1 - (1+h)\} \{2 - (1+h)\} - 0}{h} = -1$$

$\therefore q = p \therefore$ Differentiable at $x = 1$.

$\Rightarrow$ Continuous at $x = 1$.

Check the differentiability at $x = 2$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-2+h)(2-2+h) - 0}{-h} = 1$$

$$= \text{finite}$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(3-2-h) - 0}{h} \rightarrow \infty \text{ (not finite)}$$

$\therefore q \neq p \therefore$ not differentiable at $x = 2$.

Now we have to check the continuity at $x = 2$

$$\text{LHL} = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (1-x)(2-x)$$

$$= \lim_{h \rightarrow 0} (1 - (2-h))(2 - (2-h)) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3-x)$$

$$= \lim_{h \rightarrow 0} (3 - (2+h)) = 1$$

$\therefore \text{LHL} \neq \text{RHL}$

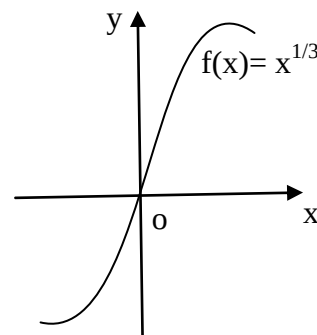
$\Rightarrow$ not continuous at $x = 2$.

Do yourself - 2:

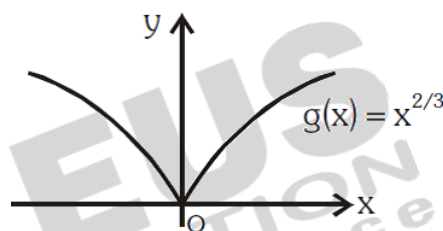
- (i) Let $f(x) = (x-1)|x-1|$. Discuss the continuity and differentiability of $f(x)$ at $x = 1$.

(b) Vertical tangent :

- (i) If for $y = f(x)$; $f'(a^+) \rightarrow \infty$ and $f'(a^-) \rightarrow -\infty$ or $f'(a^+) \rightarrow -\infty$ and $f'(a^-) \rightarrow \infty$ then at $x = a$, $y = f(x)$ has vertical tangent at $x = a$



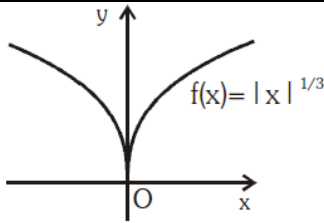
e.g. (i) $f(x) = x^{1/3}$ has vertical tangent at $x = 0$ since $f'(0^+) \rightarrow \infty$ and $f'(0^-) \rightarrow \infty$ hence $f(x)$ is not differentiable at $x = 0$



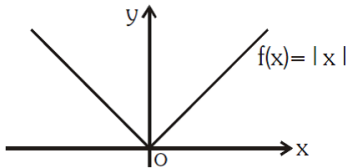
(2) $g(x) = x^{2/3}$ doesn't have vertical tangent at $x = 0$ since $g'(0^+) \rightarrow \infty$ and $g'(0^-) \rightarrow -\infty$ hence $g(x)$ is not differentiable at $x = 0$.
(ii) If a function has vertical tangent at $x = a$ then it is non differentiable at $x = 0$.

(c) Geometrical interpretation of differentiability :

- (i) If the function $y = f(x)$ is differentiable at $x = 0$, then a unique tangent can be drawn to the curve $y = f(x)$ at the point $P(a, f(a))$ and $f'(a)$ represent the slope of the tangent at point P .
- (ii) If the function $f(x)$ does not have a unique tangent ($p \neq q$) but is continuous at $x = a$. it geometrically implies a sharp corner at $x = a$. Note that p and q may not be finite, where $p = f'(a^+)$ and $q = f'(a^-)$
e.g. (1) $f(x) = |x|$ and $|x|^{1/3}$ is continuous but not differentiable at $x = 0$ and there is sharp corner at $x = 0$.

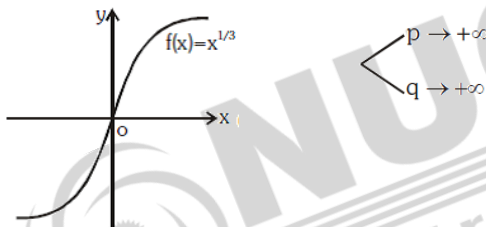


(does not have unique tangent) $\begin{cases} p \rightarrow +\infty \\ q \rightarrow -\infty \end{cases}$



(does not have unique tangent) $\begin{cases} p = 1 \\ q = -1 \end{cases}$

(2) $f(x) = x^{1/3}$ is continuous but not differentiable at $x = 0$ because $f'(0^+) \rightarrow \infty$ and $f'(0^-) \rightarrow \infty$.



(have a unique tangent but does not have sharp corner)

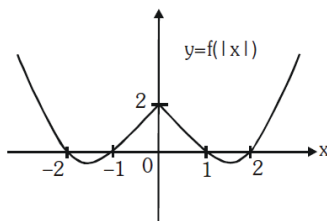
Note :

sharp corner $\Rightarrow$ non differentiable
non differentiable $\nRightarrow$ sharp corner

Illustration 5:

If $f(x) = \begin{cases} x-3 & x < 0 \\ x^2-3x+2 & x \geq 0 \end{cases}$. Draw the graph of the function and discuss the continuity and differentiability of $f(|x|)$ and $|f(x)|$.

Solution :



$$f(|x|) = \begin{cases} |x| - 3; & |x| < 0 \rightarrow \text{not possible} \\ |x|^2 - 3|x| + 2; & |x| \geq 0 \end{cases}$$

$$f(|x|) = \begin{cases} x^2 + 3x + 2, & x < 0 \\ x^2 - 3x + 2, & x \geq 0 \end{cases}$$

at $x = 0$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{-h} = 3$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3$$

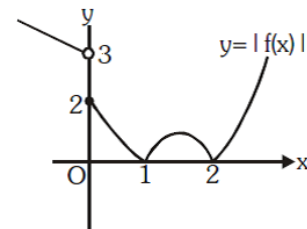
$\therefore q \neq p$

$\therefore$ not differentiable at $x = 0$. but p and q both are finite

$\Rightarrow$ continuous at $x = 0$

Now,

$$|f(x)| = \begin{cases} 3-x & , x < 0 \\ (x^2-3x+2) & , 0 \leq x < 1 \\ -(x^2-3x+2) & , 1 \leq x \leq 2 \\ (x^2-3x+2) & , x > 2 \end{cases}$$



To check differentiability at $x = 0$

$$\left. \begin{aligned} q = \text{LHD} &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{3+h-2}{-h} = \lim_{h \rightarrow 0} \frac{(1+h)}{-h} \rightarrow -\infty \\ p = \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3 \end{aligned} \right\} \Rightarrow \text{not differentiable at } x = 0.$$

Now to check continuity at $x = 0$

$$\left. \begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 3 - x = 3 \\ \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^2 - 3x + 2 = 2 \end{aligned} \right\}$$

$\Rightarrow$ not continuous at $x = 0$.

To check differentiability at $x = 1$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h^2) - 3(1-h) + 2 - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + h}{-h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-(h^2 + 2h + 1 - 3 + 3h + 2) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-(h^2 - h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 1$.

but $|f(x)|$ is continuous at $x = 1$, because

$p \neq q$ and both are finite.

To check differentiability at $x = 2$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-(4 + h^2 - 4h - 6 + 3h + 2) - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - h}{h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(h^2 + 4h + 4 - 6 - 3h + 2) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(h^2 + h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 2$.

but $|f(x)|$ is continuous at $x = 2$, because

$p \neq q$ and both are finite.

Do yourself - 3:

(i) Let $f(x) = \begin{cases} -4 & ; \quad -4 < x < 0 \\ x^2 - 4 & ; \quad 0 \leq x < 4 \end{cases}$

Discuss the continuity and differentiability of $g(x) = |f(x)|$.

(ii) Let $f(x) = \min \{|x-1|, |x+1|, 1\}$. Find the number of points where it is not differentiable.

5. DIFFERENTIABILITY OVER AN INTERVAL :

(a) $f(x)$ is said to be differentiable over an open interval (a, b) if it is differentiable at each and every point of the open interval (a, b) .

(b) $f(x)$ is said to be differentiable over the closed interval $[a, b]$ if :

(i) $f(x)$ is differentiable in (a, b) and

(ii) for the points a and b , $f'(a^+)$ and $f'(b^-)$ exist.

Illustration 6:

If

$$f(x) = \begin{cases} e^{-|x|}, & -5 < x < 0 \\ -e^{-|x-1|} + e^{-1} + 1, & 0 \leq x < 2 \\ e^{-|x-2|}, & 2 \leq x < 4 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ in the interval $(-5, 4)$.

Solution :

$$f(x) = \begin{cases} e^{+x} & -5 < x < 0 \\ -e^{x-1} + e^{-1} + 1 & 0 \leq x \leq 1 \\ -e^{-x+1}e^{-1} + 1 & 1 < x < 2 \\ e^{-x+2} & 2 \leq x < 4 \end{cases}$$

Check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-e^{h-1} + e^{-1} + 1 - 1}{h} = -e^{-1}$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 0$, but continuous at $x = 0$ since LHD and RHD both are finite.

Check the differentiability at $x = 1$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{-h} = -1$$

$$\begin{aligned} \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{h} = 1 \end{aligned}$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 1$, but continuous at $x = 1$ since LHD and RHD both are finite.

Check the differentiability at $x = 2$

$$\begin{aligned} \text{LHD} &= \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-e^{-2+h+1} + e^{-1} + 1 - 1}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-e^{-1}(e^h - 1)}{-h} = e^{-1} \end{aligned}$$

$$\begin{aligned} \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{h} = -1 \end{aligned}$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 2$, but continuous at $x = 2$ since LHD and RHD both are finite.

Note :

- (i) If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$, then the product function $F(x) = f(x) \cdot g(x)$ can still be differentiable at $x = a$.
e.g. Consider $f(x) = x$ and $g(x) = |x|$, f is differentiable at $x = 0$ and g is non-differentiable at $x = 0$, but $f(x) \cdot g(x)$ is still differentiable at $x = 0$.
- (ii) If $f(x)$ and $g(x)$ both are not differentiable at $x = a$ then the product function ; $F(x) = f(x) \cdot g(x)$ can still be differentiable at $x = a$.
e.g. Consider $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non differentiable at $x = 0$, but $f(x) \cdot g(x)$ still differentiable at $x = 0$.
- (iii) If $f(x)$ & $g(x)$ both are non-differentiable at $x = a$ then the sum function $F(x) = f(x) + g(x)$ may be a differentiable function.

e.g. $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non differentiable at $x = 0$, but $(f + g)(x)$ still differentiable at $x = 0$.

- (iv) If $f(x)$ is differentiable at $x = a \Rightarrow f'(x)$ is continuous at $x = a$.

$$\text{e.g. } f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Do yourself - 4 :

- (i) Let $f(x) = \max \{ \sin x, 1/2 \}$, where $0 \leq x \leq \frac{5\pi}{2}$. Find the number of points where it is not differentiable.
- (ii) Let $f(x) = \begin{cases} [x] & ; 0 < x \leq 2 \\ 2x - 2 & ; 2 < x < 3 \end{cases}$, where $[.]$ denotes greatest integer function.
(a) Find that points at which continuity and differentiability should be checked.
(b) Discuss the continuity and differentiability of $f(x)$ in the interval $(0, 3)$.

6. DETERMINATION OF FUNCTION WHICH SATISFYING THE GIVEN FUNCTIONAL RULE :

Illustration 7:

Let $f(x + y) = f(x) + f(y) - 2xy - 1$ for all x and y . If $f'(0)$ exists and $f'(0) = -\sin \alpha$, then find $f\{f'(0)\}$.

Solution :

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\{f(x) + f(h) - 2xy - 1\} - f(x)}{h} \\ &\text{(Using the given relation)} \\ &= \lim_{h \rightarrow 0} -2x + \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \\ &= -2x + \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \end{aligned}$$

[Putting $x = 0 = y$ in the given relation we find $f(0) = f(0) + f(0) + 0 - 1 \Rightarrow f(0) = 1$]

$$\therefore f'(x) = -2x + f'(0) = -2x - \sin \alpha$$

$$\begin{aligned}\Rightarrow f(x) &= -x^2 - (\sin \alpha) \cdot x + c \\ f(0) &= -0 - 0 + c \Rightarrow c = 1 \\ \therefore f(0) &= -x^2 - (\sin \alpha) \cdot x + 1 \\ \text{So, } f\{f'(0)\} &= f(-\sin \alpha) = -\sin^2 \alpha + \sin^2 \alpha + 1 \\ \therefore f\{f'(0)\} &= 1.\end{aligned}$$

Do yourself - 5:

- (i) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$, $f(x) \neq 0$. Suppose that the function is differentiable everywhere and $f'(0) = 2$. Prove that $f'(x) = 2f(x)$.

Miscellaneous Illustration :

Illustration 8:

Discuss the continuity and differentiability of the function $y = f(x)$ defined parametrically ; $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.

Solution :

Here $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.
Now when $t < 0$;
 $x = 2t - \{-(t - 1)\} = 3t - 1$ and $y = 2t^2 - t^2 = t^2 \Rightarrow y = \frac{1}{9}(x + 1)^2$
when $0 \leq t < 1$
 $x = 2t - (-(t - 1)) = 3t - 1$ and $y = 2t^2 + t^2 = 3t^2 \Rightarrow y = \frac{1}{3}(x + 1)^2$
when $t \geq 1$;
 $x = 2t - (t - 1) = t + 1$ and $y = 2t^2 + t^2 = 3t^2 \Rightarrow y = 3(x - 1)^2$
Thus,

$$y = f(x) = \begin{cases} \frac{1}{9}(x+1)^2, & x < -1 \\ \frac{1}{3}(x+1)^2, & -1 \leq x < 2 \\ 3(x-1)^2, & x \geq 2 \end{cases}$$

We have to check differentiability at $x = -1$ and 2 .

Differentiability at $x = -1$;

$$\text{LHD} = f'(-1^-) = \lim_{h \rightarrow 0} \frac{f(-1-h) - f(-1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{9}(-1-h+1)^2 - 0}{-h} = 0$$

$$\text{RHD} = f'(-1^+) = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{3}(-1+h+1)^2 - 0}{h} = 0$$

Hence $f(x)$ is differentiable at $x = -1$.

$\Rightarrow$ continuous at $x = -1$.

To check differentiability at $x = 2$;

$$\text{LHD} = f'(2^-) = \lim_{h \rightarrow 0} \frac{\frac{1}{3}(2-h-1)^2 - 3}{-h} = 2$$

$$\text{and RHD} = f'(2^+) = \lim_{h \rightarrow 0} \frac{3(2+h-1)^2 - 3}{h} = 6$$

Hence $f(x)$ is not differentiable at $x = 2$.
But continuous at $x = 2$, because LHD and RHD both are finite.

$\therefore f(x)$ is continuous for all x and differentiable for all x , except $x = 2$.

ANSWERS FOR DO YOURSELF

- (i) Continuous but not differentiable at $x = 1$
(ii) $a = \frac{1}{6}$, $b = \frac{\pi}{4} - \frac{13}{6}$
- (i) Continuous and differentiable at $x = 1$
- (i) Continuous everywhere but not differentiable at $x = 2$ only
(ii) 5
- (i) 3
(ii) (a) 1 and 2
(b) Not continuous at $x = 1$ and 2 and not differentiable at $x = 1$ and 2 .

Result JEE Main

Result Highlights 2019

							
100 Percentile HIMANSHU GAURAV SINGH 2019 (*SDCCP)	99.99 Percentile GAURAV KRISHAN GUPTA 2020 (DLP)	99.98 Percentile SARTHAK ROUT 2020 (CCP)	99.98 Percentile VIBHAV AGGARWAL 2019 (CCP)	99.97 Percentile RITVIK GUPTA 2020 (DLP)	99.97 Percentile BHAVYA JAIN 2020 (CCP)	99.96 Percentile AYUSH PATTNAIK 2019 (CCP)	99.96 Percentile SAYANTAN DHAR 2020 (DLP)

Result Highlights 2018

AIR 29 	AIR 36 	AIR 46 	AIR 48 	AIR 2 SC 	AIR 18 OBC 
Sarthak Behera	Shashank Agrawal	Shreya Pathak	Raaghav Raaj	Aayush Kadam	Gunjan

Result Highlights 2017

AIR 12 	AIR 14 	AIR 20 	AIR 38 	AIR 49 	AIR 60 
Abhay Goyal	Piyush Tibarwal	Abhinav Kumar	Tushar Gautam	Saurav Sahoo	Pratyush

Mentored by Team

AIR 4 	AIR 15 	AIR 21 	AIR 24 	AIR 38 
Kushal Babel 2014	Gourika Bang 2015	Amit Malav 2013	Suraj Sanjay Jog 2012	Chitraang Murdia 2014
AIR 58 	AIR 68 	AIR 70 	AIR 80 	
Pulkit Tandon 2012	Kapil Shobhnani 2016	Utkarsh Sinha 2015	S John S 2015	

IIT MATHEMATICS

LIMIT, CONTINUITY & DIFFERENTIABILITY



RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
Gen. Category - 2014

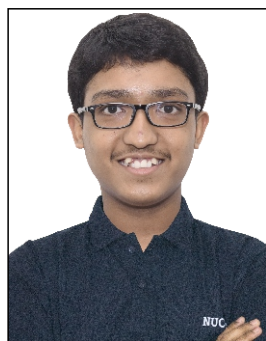


AIR-6
NISHIT AGARWAL
Gen. Category - 2012

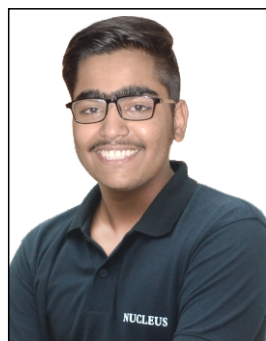


AIR-8
AMEY GUPTA
Gen. Category - 2014

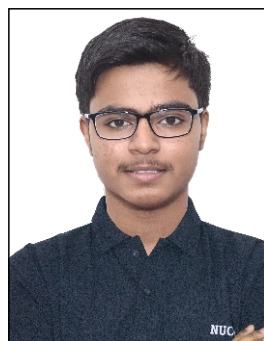
JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



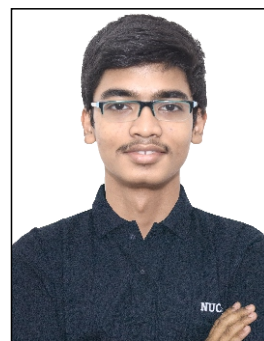
100 Percentile
(Physics)
DAKSH Khandelwal
2020



100 Percentile
(Maths & Physics)
VAIBHAV SAHA
2020



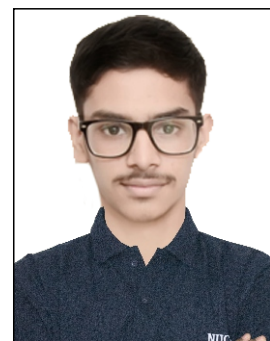
100 Percentile
(Physics)
ANISH MOHAN
2020



100 Percentile
(Maths)
ARCHIT PATNAIK
2020

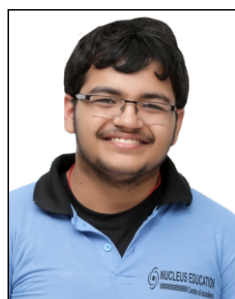


100 Percentile
(Maths)
SWAPNIL YASASVI
2020

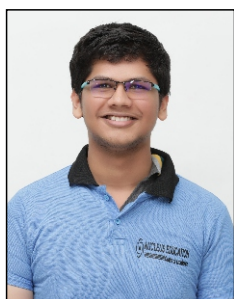


100 Percentile
(Maths)
PARSHANT ARORA
2020

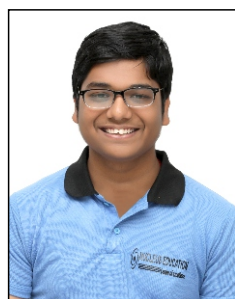
JEE ADVANCED RESULTS OF NUCLEUS EDUCATION



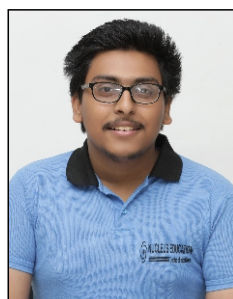
AIR-10
LAKSHAY SHARMA
Gen. Category - 2017



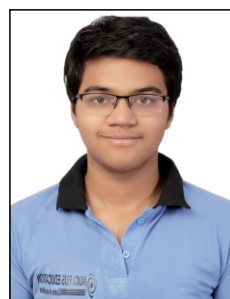
AIR-12
YATEESH AGRAWAL
Gen. Category - 2017



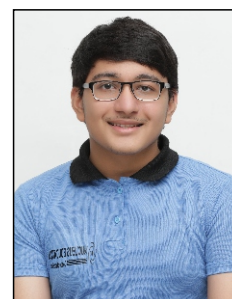
AIR-23
ABHEY GOYAL
Gen. Category - 2017



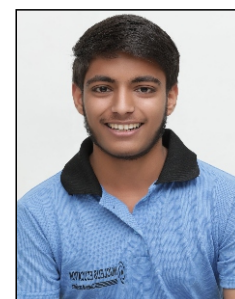
AIR-24
TUSHAR GAUTAM
Gen. Category - 2017



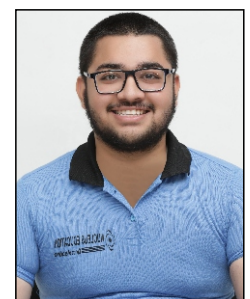
AIR-37
PIYUSH TIBAREWAL
Gen. Category - 2017



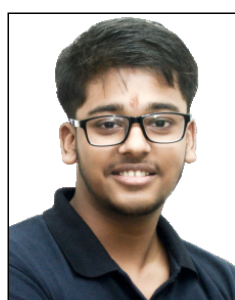
AIR-42
SATVIK
Gen. Category - 2017



AIR-66
MAYANK DUBEY
Gen. Category - 2017



AIR-98
HRITHIK
Gen. Category - 2017



AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



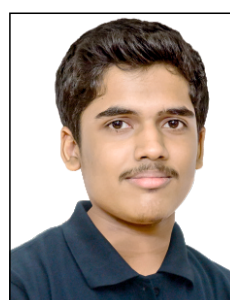
AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
Gen. Category - 2018



AIR-61
SIDDHANT CHOUDHARY
Gen. Category - 2018



AIR-67
ANISWAR S KRISHNAN
Gen. Category - 2018
DLP



AIR-78
AAYUSH KADAM
Gen. Category - 2018



AIR-61
SARTHAK BEHERA
Gen. Category - 2018
*SDCCP



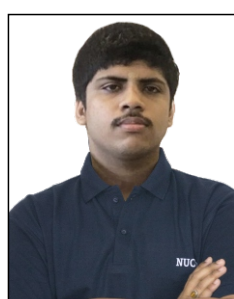
AIR-91
ANDREWS G. VARGHESE
Gen. Category - 2018
*SDCCP



AIR-2
HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



AIR-19
VIBHAV AGGARWAL
Gen. Category - 2019



AIR-33
S. PRAJEETH
Gen. Category - 2019
*SDCCP



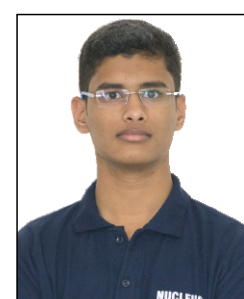
AIR-48
SOHAM MISTRI
Gen. Category - 2019
DLP



AIR-51
SAYANTAN DHAR
Gen. Category - 2019
DLP



AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
DLP



AIR-86
SATVIK JAIN
Gen. Category - 2019

LIMIT

EXERCISE # 1

[SINGLE CORRECT CHOICE TYPE]

1. $\lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$ is equal to
 (A) -1 (B) 0 (C) 1 (D) D.N.E.
2. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$ is equal to
 (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
3. $\lim_{x \rightarrow 2} \frac{\sqrt{1+\sqrt{2+x}} - \sqrt{3}}{x-2}$ is equal to
 (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{1}{4\sqrt{3}}$ (D) $\frac{1}{8\sqrt{3}}$
4. $\lim_{x \rightarrow 1} \frac{\sqrt[n]{x} - 1}{\sqrt[m]{x} - 1}$ (m and n integers) is equal to
 (A) 0 (B) 1 (C) $\frac{m}{n}$ (D) $\frac{n}{m}$
5. If $\lim_{x \rightarrow a} \frac{2x - \sqrt{x^2 + 3a^2}}{\sqrt{x+a} - \sqrt{2a}} = \sqrt{2}$ (where $a \in \mathbb{R}^+$), then a is equal to –
 (A) $\frac{1}{3}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{9}$
6. $\lim_{x \rightarrow 0} \frac{\ln(\sin 3x)}{\ln(\sin x)}$ is equal to
 (A) 0 (B) 1 (C) 2 (D) Non existent
7. $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$ is equal to
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) D.N.E.
8. $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^3} - \sqrt{3+x^2}}{x-1}$ is equal to
 (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{6}$

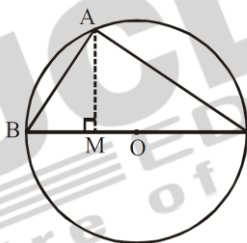
9. $\lim_{n \rightarrow \infty} \frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$ is equal to
(A) -1 (B) 0 (C) 1 (D) D.N.E.
10. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to
(A) 1 (B) 100 (C) 200 (D) 10
11. $\lim_{x \rightarrow -\infty} (\sqrt{x^2 - 2x - 1} - \sqrt{x^2 - 7x + 3})$ is equal to :
(A) $-\frac{5}{2}$ (B) $\frac{5}{2}$ (C) 0 (D) D.N.E.
12. If $\lim_{n \rightarrow \infty} (\sqrt{2n^2 + n} - \lambda \sqrt{2n^2 - n}) = \frac{1}{\sqrt{2}}$ (where λ is a real number), then -
(A) $\lambda = 1$ (B) $\lambda = -1$ (C) $\lambda = \pm 1$ (D) $\lambda \in (-\infty, 1)$
13. Let $U_n = \frac{n!}{(n+2)!}$ where $n \in \mathbb{N}$. If $S_n = \sum_{n=1}^n U_n$ then $\lim_{n \rightarrow \infty} S_n$ equals
(A) 2 (B) 1 (C) 1/2 (D) Non existent
14. For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$. Then $\lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n})$ is equal to -
(A) 1 (B) $\frac{1}{2}$ (C) 0 (D) 2
15. Let $P_n = \prod_{k=2}^n \left(1 - \frac{1}{k+1} C_2\right)$. If $\lim_{n \rightarrow \infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a+b)$ is:
(A) 4 (B) 8 (C) 10 (D) 12
16. $\lim_{x \rightarrow -1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to
(A) 0 (B) $\cos 2$ (C) $2\sin 2$ (D) $\sin 1$
17. Let $f(x) = \left[\frac{\sin x}{x} \right] + \left[\frac{2\sin 2x}{x} \right] + \dots + \left[\frac{10\sin 10x}{x} \right]$ (where $[y]$ is the largest integer $\leq y$). The value of $\lim_{x \rightarrow 0} f(x)$ equals :
(A) 55 (B) 164 (C) 165 (D) 375

18. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ and $f(3^+)$ exists finitely and are not zero, then the value of $(a + b)$ is (where $\{.\}$ represents fractional part function) -
(A) 7 (B) 10 (C) 11 (D) 20

19. $\lim_{x \rightarrow 0} \frac{|\cos(\sin(3x))| - 1}{x^2}$ equals
(A) $\frac{-9}{2}$ (B) $\frac{-3}{2}$ (C) $\frac{3}{2}$ (D) $\frac{9}{2}$

20. Let $a = \min \{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$. then value of $\sum_{r=0}^n a^r \cdot b^{n-r}$ is :
(A) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ (B) $\frac{2^{n+1} + 1}{3 \cdot 2^n}$ (C) $\frac{4^{n+1} - 1}{3 \cdot 2^n}$ (D) none of these

21. Let BC is diameter of a circle centred at O. Point A is a variable point, moving on the circumference of circle. if $BC = 1$ unit, then $\lim_{A \rightarrow B} \frac{BM}{(\text{Area of sector OAB})^2}$ is equal to -



- (A) 1 (B) 2 (C) 4 (D) 16

22. $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 2} \right)^x$ is equal to

- (A) 1 (B) e (C) $\frac{1}{e^2}$ (D) e^2

23. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cos x}$ is equal to

- (A) 0 (B) e (C) 1 (D) $\frac{1}{e}$

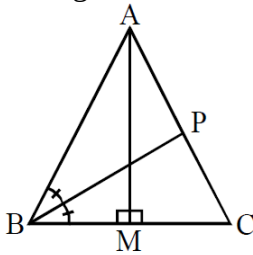
24. $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{1/x}$ is equal to :

- (A) e^a (B) e^{ab} (C) e^b (D) $e^{a/b}$

25. $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x \right) \right)^{1/x}$ is equal to

- (A) e^{-2} (B) $\frac{1}{e}$ (C) e (D) e^2

26. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to
 (A) 5 (B) 4 (C) 0 (D) D.N.E.
27. $\lim_{x \rightarrow \infty} \left(\frac{1^{1/x} + 2^{1/x} + 3^{1/x} + \dots + n^{1/x}}{n} \right)^{nx}$ is equal to
 (A) $n!$ (B) 1 (C) $\frac{1}{n!}$ (D) 0
28. If $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x} = e^3$, then
 (A) $a = \frac{3}{2}$ and $b \in \mathbb{R}$ (B) $a = \frac{3}{2}$ and $b \in \mathbb{R}^+$
 (C) $a = 0$ and $b = 1$ (D) $a = 1$ and $b = 0$
29. If $f(x)$ is a polynomial of least degree, such that $\lim_{x \rightarrow 0} \left(1 + \frac{f(x) + x^2}{x^2} \right)^{1/x} = e^2$, then $f(2)$ is -
 (A) 2 (B) 8 (C) 10 (D) 12
30. $\lim_{n \rightarrow \infty} \frac{e^n}{\left(1 + \frac{1}{n}\right)^{n^2}}$ equals -
 (A) 1 (B) $\frac{1}{2}$ (C) e (D) $\sqrt{e}$
31. If $f(x)$ is odd linear polynomial with $f(1) = 1$, then $\lim_{x \rightarrow 0} \frac{2^{f(\tan x)} - 2^{f(\sin x)}}{x^2 f(\sin x)}$ is :
 (A) 1 (B) $\ln 2$ (C) $\frac{1}{2} \ln 2$ (D) $\cos 2$
32. $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = 1$ then
 (A) $a = -5/2$ (B) $a = -3/2, b = -1/2$
 (C) $a = -3/2, b = -5/2$ (D) $a = -5/2, b = -3/2$
33. $\lim_{h \rightarrow 0} \frac{\sin(a+3h) - 3\sin(a+2h) + 3\sin(a+h) - \sin a}{h^3}$ is equal to
 (A) $\cos a$ (B) $-\cos a$ (C) $\sin a$ (D) $\sin a \cos a$

34. $\lim_{x \rightarrow \frac{\pi}{2}} \tan^2 x \left(\sqrt{2 \sin^2 x + 3 \sin x + 4} - \sqrt{\sin^2 x + 6 \sin x + 2} \right)$ is equal to
 (A) $\frac{3}{4}$ (B) $\frac{1}{6}$ (C) $\frac{1}{12}$ (D) $\frac{5}{12}$
35. $\lim_{x \rightarrow \infty} x \left(\arctan \frac{x+1}{x+2} - \arctan \frac{x}{x+2} \right)$ is equal to
 (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) D.N.E.
36. $\lim_{h \rightarrow 0} \frac{\tan(a+2h) - 2 \tan(a+h) + \tan a}{h^2}$ is equal to
 (A) $\tan a$ (B) $\tan^2 a$ (C) $\sec a$ (D) $2(\sec^2 a)(\tan a)$
37. $\lim_{x \rightarrow 0} \left(2^{x-1} + \frac{1}{2} \right)^{1/x}$ equals
 (A) $\sqrt{2}$ (B) $\frac{1}{2} \ln 2$ (C) $\ln 2$ (D) 2
38. If $\lim_{x \rightarrow 0} \left(\cos x + a^3 \sin(b^6 x) \right)^{\frac{1}{x}} = e^{512}$, then the value of ab^2 is equal to
 (A) -512 (B) 512 (C) 8 (D) $8\sqrt{8}$
39. The value of $\lim_{x \rightarrow 0} \frac{\sin(\sqrt[3]{x}) \ln(1+3x)}{\left(\tan^{-1} \sqrt{x} \right)^2 \left(e^{5\sqrt[3]{x}} - 1 \right)}$ is equal to
 (A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{4}{5}$
40. The figure shows an isosceles triangle ABC with $\angle B = \angle C$. The bisector of angle B intersects the side AC at the point P. Suppose that BC remains fixed but the altitude AM approaches 0, so that $A \rightarrow M$ (mid-point of BC). Limiting value of BP, is

 (A) $\frac{a}{3}$ (B) $\frac{a}{2}$ (C) $\frac{2a}{3}$ (D) $\frac{3a}{4}$
 where a is fixed side BC.

41. The value of $\lim_{x \rightarrow 2} \frac{\sec^x \theta - \tan^x \theta - 1}{x - 2}$ is equal to-
- (A) $\sec^2 \theta \cdot \ln \sec \theta + \tan^2 \theta \cdot \ln \tan \theta$ (B) $\sec^2 \theta \cdot \ln \tan \theta + \tan^2 \theta \cdot \ln \sec \theta$
 (C) $\sec^2 \theta \cdot \ln \tan \theta - \tan^2 \theta \cdot \ln \sec \theta$ (D) $\sec^2 \theta \cdot \ln \sec \theta - \tan^2 \theta \cdot \ln \tan \theta$
42. Consider the function $f(x) = \begin{cases} 1-x, & 0 \leq x \leq 1 \\ x+2, & 1 < x < 2 \\ 4-x, & 2 \leq x \leq 4 \end{cases}$. Let $\lim_{x \rightarrow 1} f(f(x)) = \ell$ and $\lim_{x \rightarrow 2} f(f(x)) = m$ then which one of the following hold good ?
- (A) ℓ exists but m does not. (B) m exists but ℓ does not.
 (C) Both ℓ and m exist (D) Neither ℓ nor m exist.
43. Let $f(x)$ be a quadratic function such that $f(0) = f(1) = 0$ and $f(2) = 1$, then $\lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} \cos^2 x\right)}{f^2(x)}$ is equal to-
- (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

EXERCISE # 2

1. $\lim_{x \rightarrow 1} \frac{x^2 - x \cdot \ln x + \ln x - 1}{x - 1}$
2. $\lim_{x \rightarrow 1} \frac{\left[\sum_{k=1}^{100} x^k \right] - 100}{x - 1}$
3. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$
4. $\lim_{x \rightarrow 0} \frac{8}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$
5. $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$
6. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin \frac{\pi}{3}}{h^4}$
7. $\lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt{\frac{x+3}{x}} \right)$
8. $\lim_{x \rightarrow -\infty} \frac{(3x^4 + 2x^2) \sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$
9. If $\ell = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$ then find $\{\ell\}$.
(where $\{ \}$ denotes the fractional part function)
10. Find a and b if :
 - (i) $\lim_{x \rightarrow \infty} \left[\frac{x^2 + 1}{x + 1} - ax - b \right] = 0$
 - (ii) $\lim_{x \rightarrow -\infty} \left[\sqrt{x^2 - x + 1} - ax - b \right] = 0$
11. $\lim_{x \rightarrow 0} [\ln(1 + \sin^2 x) \cdot \cot(\ell n^2(1 + x))]$

12. $\lim_{x \rightarrow 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$

13. (a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$, where $a \in \mathbb{R}$;

(b) Plot the graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$

14. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that

(i) $a_n + b_n + c_n = 2n + 1$;

(ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$;

(iii) $a_n b_n c_n = -1$;

(iv) $a_n < b_n < c_n$

Then find the value of $\lim_{n \rightarrow \infty} (na_n)$.

15. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.

If $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ and $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$, then find the value of $\frac{c^2 + d^2}{a^2 + b^2}$.

16. $\lim_{x \rightarrow \infty} \left[\frac{2x^2 + 3}{2x^2 + 5} \right]^{8x^2 + 3}$

17. $\lim_{x \rightarrow \infty} \left(\frac{x+c}{x-c} \right)^x = 4$ then find c

18. $\lim_{x \rightarrow 1} \left(\tan \frac{\pi x}{4} \right)^{\tan \frac{\pi x}{2}}$

19. $\lim_{x \rightarrow 0} \left(\frac{x-1+\cos x}{x} \right)^{\frac{1}{x}}$

20. If $n \in \mathbb{N}$ and $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$.

Find the value $\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}$.

EXERCISE # 3 (JM)

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$ [AIEEE-2010]
 (1) 1 (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) 3
- $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$ [AIEEE-2011]
 (1) equals $-\sqrt{2}$ (2) equals $\frac{1}{\sqrt{2}}$ (3) does not exist (4) equals $\sqrt{2}$
- Let $f : \mathbb{R} \rightarrow [0, \infty)$ be such that $\lim_{x \rightarrow 5} f(x)$ exists and $\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$ Then $\lim_{x \rightarrow 5} f(x)$ equal - [AIEEE-2011]
 (1) 3 (2) 0 (3) 1 (4) 2
- $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ is equal to : [JEE Mains Offline-2014]
 (1) $\frac{\pi}{2}$ (2) 1 (3) $-\pi$ (4) π
- If $\lim_{x \rightarrow 2} \frac{\tan(x-2)\{x^2 + (k-2)x - 2k\}}{x^2 - 4x + 4} = 5$ then k is equal to [JEE Mains Online-2014]
 (1) 3 (2) 1 (3) 0 (4) 2
- Let $p = \lim_{x \rightarrow 0^+} \left(1 + \tan^2 \sqrt{x}\right)^{\frac{1}{2x}}$ then $\log p$ is equal to - [JEE(Main)-2016]
 (1) $\frac{1}{4}$ (2) 2 (3) 1 (4) $\frac{1}{2}$
- $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x - \cos x}{(\pi - 2x)^3}$ equals : [JEE(Main)-2017]
 (1) $\frac{1}{4}$ (2) $\frac{1}{24}$ (3) $\frac{1}{16}$ (4) $\frac{1}{8}$
- For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then
 $\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$ [JEE(Main)-2018]
 (1) does not exist (in $\mathbb{R}$) (2) is equal to 0. (3) is equal to 15. (4) is equal to 120.

9. $\lim_{y \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4}$ [JEE(Main) Online -2019]
 (1) exists and equals $\frac{1}{4\sqrt{2}}$ (2) does not exist
 (3) exists and equals $\frac{1}{2\sqrt{2}}$ (4) exists and equals $\frac{1}{2\sqrt{2}(\sqrt{2} + 1)}$
10. For each $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . Then
 $\lim_{x \rightarrow 0^-} \frac{x([x] + |x|)\sin[x]}{|x|}$ [JEE(Main) Online -2019]
 (1) $-\sin 1$ (2) $\sin 1$ (3) 1 (4) 0
11. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then,
 $\lim_{x \rightarrow 1^+} \frac{(1 - |x| + \sin |1 - x|) \sin\left(\frac{\pi}{2}[1 - x]\right)}{|1 - x|[1 - x]}$ [JEE(Main) Online -2019]
 (1) equals 0 (2) does not exist (3) equals -1 (4) equals 1
12. Let $[x]$ denote the greatest integer less than or equal to x .
 Then : $\lim_{x \rightarrow 0} \frac{\tan(\pi \sin^2 x) + (|x| - \sin(x[x]))^2}{x^2}$ [JEE(Main) Online -2019]
 (1) does not exist (2) equals π (3) equals 0 (4) equals $\pi + 1$
13. $\lim_{x \rightarrow 0} \frac{x \cot(4x)}{\sin^2 x \cot^2(2x)}$ is equal to: [JEE(Main) Online -2019]
 (1) 1 (2) 4 (3) 0 (4) 2
14. $\lim_{x \rightarrow \pi/4} \frac{\cot^3 x - \tan x}{\cos\left(x + \frac{\pi}{4}\right)}$ is : [JEE(Main) Online -2019]
 (1) $4\sqrt{2}$ (2) 8 (3) 4 (4) $8\sqrt{2}$
15. $\lim_{x \rightarrow 1^-} \frac{\sqrt{\pi} - \sqrt{2 \sin^{-1} x}}{\sqrt{1 - x}}$ is equal to: [JEE(Main) Online -2019]
 (1) $\sqrt{\pi}$ (2) $\sqrt{\frac{\pi}{2}}$ (3) $\sqrt{\frac{2}{\pi}}$ (4) $\frac{1}{\sqrt{2\pi}}$
16. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equals [JEE(Main) Online -2019]
 (1) $2\sqrt{2}$ (2) 4 (3) $4\sqrt{2}$ (4) $\sqrt{2}$

17. If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is : [JEE(Main) Online -2019]
 (1) $\frac{3}{2}$ (2) $\frac{4}{3}$ (3) $\frac{3}{8}$ (4) $\frac{8}{3}$
18. If $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$, then a + b is equal to : [JEE(Main) Online -2019]
 (1) -7 (2) 5 (3) -4 (4) 1
19. $\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$ is : [JEE(Main) Online -2019]
 (1) 3 (2) 2 (3) 1 (4) 6
20. Let $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$, $x \in \mathbb{R}$. If $f(x)$ attains maximum value at α and $g(x)$ attains minimum value at β , then $\lim_{x \rightarrow -\alpha\beta} \frac{(x-1)(x^2-5x+6)}{x^2-6x+8}$ is equal to : [JEE(Main) Online -2019]
 (1) -3/2 (2) 3/2 (3) 1/2 (4) -1/2
21. $\lim_{x \rightarrow 2} \frac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$ is equal to _____. [JEE(Main) Online -2020]
22. $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{1/x^2}$ is equal to [JEE(Main) Online -2020]
 (1) e^2 (2) e (3) $\frac{1}{e}$ (4) $\frac{1}{e^2}$

EXERCISE # 4 (JA)

SECTION-1

1. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1+b^2) \right]^{\frac{1}{x}} = 2 b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is-
 (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$
[JEE 2012, 3M, -1M]
2. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then -
 (A) $a = 1, b = 4$ (B) $a = 1, b = -4$ (C) $a = 2, b = -3$ (D) $a = 2, b = 3$
3. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$ where $a > -1$. then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are **[JEE 2012, 3m, -1M]**
 (A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3

SECTION-2

4. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then - **[JEE 2009, 4]**
 (A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$
5. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is **[JEE (Advanced)-2014, 3]**
6. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals **[JEE (Advanced)-2016, 3(0)]**
7. Let $f(x) = \frac{1 - x(1 + |1 - x|)}{|1 - x|} \cos\left(\frac{1}{1 - x}\right)$ for $x \neq 1$, Then
 (A) $\lim_{x \rightarrow 1^+} f(x)$ does not exist
 (B) $\lim_{x \rightarrow 1^-} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^+} f(x) = 0$
 (D) $\lim_{x \rightarrow 1^-} f(x) = 0$

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say that f has

[JEE (Advanced)-2019]

PROPERTY 1 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite, and

PROPERTY 2 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct ?

(A) $f(x) = x^{2/3}$ has PROPERTY 1

(B) $f(x) = \sin x$ has PROPERTY 2

(C) $f(x) = x|x|$ has PROPERTY 2

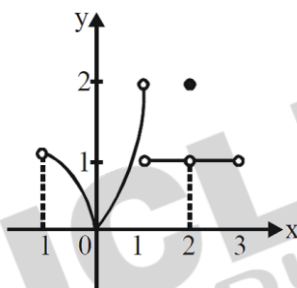
(D) $f(x) = |x|$ has PROPERTY 1



EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- If $\ell = \lim_{x \rightarrow a} \frac{\sqrt{3x^2 + a^2} - \sqrt{x^2 + 3a^2}}{(x-a)}$ then -
 (A) $\ell = 1 \forall a \in \mathbb{R}$ (B) $\ell = 1 \forall a > 0$ (C) $\ell = -1 \forall a < 0$ (D) $\ell = \text{D.N.E}$ if $a = 0$
- Which of the following limits vanish ?
 (A) $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$ (B) $\lim_{x \rightarrow \infty} \frac{\arctan x}{x}$ (C) $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x + \cos x}$ (D) $\lim_{x \rightarrow 1} \frac{\arcsin x}{\tan \frac{\pi x}{2}}$
- Which of the following statement are true for the function f defined for $-1 \leq x \leq 3$ in the figure shown .



- (A) $\lim_{x \rightarrow -1^+} f(x) = 1$ (B) $\lim_{x \rightarrow 2} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^+} f(x) = 2$ (D) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
- Let $f(x) = x + \sqrt{x^2 + 2x}$ and $g(x) = \sqrt{x^2 + 2x} - x$, then
 (A) $\lim_{x \rightarrow \infty} g(x) = 1$ (B) $\lim_{x \rightarrow \infty} f(x) = 1$ (C) $\lim_{x \rightarrow -\infty} f(x) = -1$ (D) $\lim_{x \rightarrow -\infty} g(x) = -1$
- If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, then (where $[.]$ denotes greatest integer function)
 (A) $A = 1$ (B) A does not exist
 (C) $B = 0$ (D) $B = 1$
- Which of the following limit tends to unity ?
 (A) $\lim_{x \rightarrow 0} \frac{1 - \cos x + 2 \sin x - \sin^3 x - x^2 + 3x^4}{\tan^3 x - 6 \sin^2 x + x - 5x^3}$ (B) $\lim_{x \rightarrow \infty} \frac{x}{[x]}$
 (C) $\lim_{x \rightarrow \infty} \frac{1}{\left(\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} \right)}$ (D) $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x}}{\sqrt{x + \sqrt{x + \sqrt{x}}}} \right)$

7. Which of the following limits does not exist ?

(A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$

(B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x-1| - 2}$

(C) $\lim_{x \rightarrow 0^+} (x)^{\ln x}$

(D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

(where $[.]$ represents greatest integer function)

8. The value (s) of 'n' for which $\lim_{x \rightarrow 1} \frac{e^{x-1} - x}{(x-1)^n}$ exists is / are -

(A) 1

(B) 2

(C) 3

(D) 4

9. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{(\tan x)^{2n} + x^2}{\sin^2 x + (\tan x)^{2n}} \right); & x \neq 0 \\ 1 & ; x = 0 \end{cases}$, $n \in \mathbb{N}$. Which of the following

holds good ?

(A) $f\left(-\frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right)$

(B) $f\left(-\frac{\pi}{4}\right) = f\left(-\frac{\pi^+}{4}\right)$

(C) $f\left(\frac{\pi^-}{4}\right) = f\left(\frac{\pi^+}{4}\right)$

(D) $f(0^+) = f(0) = f(0^-)$

10. Let $f(x) = \begin{cases} \frac{\tan^2 \{x\}}{x^2 - [x]^2} & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot \{x\}} & \text{for } x < 0 \end{cases}$ where $[x]$ is the step up function and $\{x\}$ is the

fractional part function of x , then -

(A) $\lim_{x \rightarrow 0^+} f(x) = 1$ (B) $\lim_{x \rightarrow 0^-} f(x) = 1$ (C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$ (D) None

11. $\lim_{x \rightarrow c} f(x)$ does not exist when (where $[x]$ is the step up function, $\{x\}$ is fractional part function of x and $\text{sgn}(x)$ denotes signum function), then-

(A) $f(x) = [[x]] - [2x - 1]; c = 3$

(B) $f(x) = [x] - x, c = 1$

(C) $f(x) = \{x\}^2 - \{-x\}^2, c = 0$

(D) $f(x) = \frac{\tan(\text{sgn} x)}{\text{sgn} x}, c = 0$

12. Which of the following limits does not exist ?

(A) $\lim_{x \rightarrow \infty} \operatorname{cosec}^{-1} \left(\frac{x}{x+7} \right)$

(B) $\lim_{x \rightarrow 1} \sec^{-1} (\sin^{-1} x)$

(C) $\lim_{x \rightarrow 0^+} x^{\frac{1}{x}}$

(D) $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{8} + x \right) \right)^{\cot x}$

13. Which of the following statement(s) is (are) **INCORRECT** ?

(A) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(x) g(x)$ also does not exist.

(B) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(g(x))$ also does not exist.

(C) If $\lim_{x \rightarrow c} f(x)$ exists and $\lim_{x \rightarrow c} g(x)$ does not exist then $\lim_{x \rightarrow c} g(f(x))$ does not exist.

(D) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both exist then $\lim_{x \rightarrow c} f(g(x))$ and $\lim_{x \rightarrow c} g(f(x))$ also exist.

14. Consider following statements and identify correct options:

(i) $\lim_{x \rightarrow 4} \left(\frac{2x}{x-4} - \frac{8}{x-4} \right) = \lim_{x \rightarrow 4} \frac{2x}{x-4} - \lim_{x \rightarrow 4} \frac{8}{x-4}$

(ii) $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 + 5x - 6} = \frac{\lim_{x \rightarrow 1} (x^2 + 6x - 7)}{\lim_{x \rightarrow 1} (x^2 + 5x - 6)}$

(iii) $\lim_{x \rightarrow 1} \frac{x-3}{x^2 + 2x - 4} = \frac{\lim_{x \rightarrow 1} (x-3)}{\lim_{x \rightarrow 1} (x^2 + 2x - 4)}$

(iv) If $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(v) If $\lim_{x \rightarrow 5} f(x) = 0$ and $\lim_{x \rightarrow 5} g(x) = 2$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(A) Only one is true.

(B) Only two are true.

(C) Only three are false.

(D) Only two are false.

15. Which of the following limits equal to $\frac{1}{2}$

(A) $\lim_{n \rightarrow \infty} \left(\frac{1}{1.3} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$

(B) $\lim_{x \rightarrow \infty} \left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2} \right]$

(C) $\lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$

(D) $\lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$

16. Let $f(x) = \begin{cases} \sin x; & \text{where } x = \text{integer} \\ 0 & ; \text{ otherwise} \end{cases}$; $g(x) = \begin{cases} x^2 + 1 & ; x \neq 0, 2 \\ 4 & ; x = 0 \\ 5 & ; x = 2 \end{cases}$, then
- (A) $\lim_{x \rightarrow 0} g(f(x)) = 4$ (B) $\lim_{x \rightarrow 0} f(g(x)) = 0$
 (C) $\lim_{x \rightarrow 1} f(g(x)) = 0$ (D) $\lim_{x \rightarrow 1} g(f(x)) = 5$

17. If $f(x) = \begin{cases} x^2, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$, then

- (A) $\lim_{x \rightarrow 0} f(x) = 0$
 (B) $\lim_{x \rightarrow 0} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 2} f(x) = 4$
 (D) $\lim_{x \rightarrow 2} f(x)$ does not exist

18. Let $f(\beta) = \lim_{\alpha \rightarrow \beta} \frac{\sin^2 \alpha - \sin^2 \beta}{\alpha^2 - \beta^2}$, then $f\left(\frac{\pi}{4}\right)$ is greater than -

- (A) $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x}$
 (B) $\lim_{x \rightarrow \pi/2} \frac{\cot x - \cos x}{(\pi - 2x)^3}$
 (C) $\lim_{x \rightarrow \infty} (\cos \sqrt{x+1} - \cos \sqrt{x})$
 (D) $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$ where $a > 0$

19. If $\frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$ has a finite limit L as $x \rightarrow 0$, then

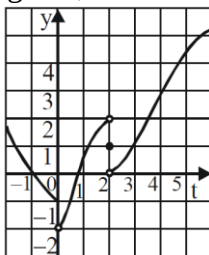
- (A) $a = -\frac{1}{2}$ (B) $b = \frac{1}{2}$ (C) $c = 0$ (D) $L = -\frac{1}{3}$

20. Let $\ell = \lim_{x \rightarrow \infty} \frac{a^x - a^{-x}}{a^x + a^{-x}}$ ($a > 0$), then

- (A) $\ell = 1 \forall a > 0$ (B) $\ell = -1 \forall a \in (0, 1)$
 (C) $\ell = 0$, if $a = 1$ (D) $\ell = 1 \forall a > 1$

[MATCH THE COLUMN TYPE]

21. For the function $g(t)$ whose graph is given, match the entries of **column-I** to **column-II**



Column-I

- (A) $\lim_{t \rightarrow 0^+} g(t) + \lim_{t \rightarrow 2^-} g(t)$
(B) $\lim_{t \rightarrow 0^-} g(t) + g(2)$
(C) $\lim_{t \rightarrow 0} g(t)$
(D) $\lim_{t \rightarrow 2} g(t)$

Column-II

- (P) $\lim_{t \rightarrow 2^+} g(t)$
(Q) does not exist
(R) 0
(S) $\lim_{t \rightarrow 4} g(t)$

22. **Column -I**

- (A) $\lim_{n \rightarrow \infty} n \sin \left(\frac{\pi}{4n} \right) \cos \left(\frac{\pi}{4n} \right)$ is equal to
(B) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ is equal to
(C) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$ is equal to
(D) $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$ is equal to

Column - II

- (P) 0
(Q) $\frac{1}{2}$
(R) $\frac{\pi}{4}$
(S) $\frac{\pi}{180}$

23. **Column -I**

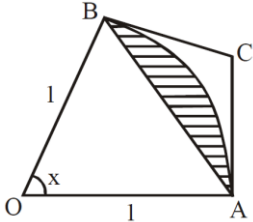
- (A) $\lim_{x \rightarrow \infty} \frac{a^x}{a^x + 1}$ ($a > 0$) can be equal to
(B) $\lim_{x \rightarrow 2} \frac{\sin(e^{x-2}-1)}{\log(x-1)}$ is equal to
(C) $\lim_{x \rightarrow e} \frac{(\ln x - 1)e}{x - e}$ is equal to
(D) $\lim_{x \rightarrow 0} \frac{x(5^x - 1)}{(1 - \cos x)4 \ln 5}$ is equal to

Column II

- (P) $\lim_{x \rightarrow \infty} x(e^{1/x} - 1)$
(Q) $\lim_{x \rightarrow 0} \frac{a^x + b^x + c^x - 3}{x}$ ($a, b, c > 0$ and $abc = 1$)
(R) $\lim_{x \rightarrow 0} \frac{e^{4x} - e^{3x}}{x}$
(S) $\frac{1}{2}$
(T) 0

EXERCISE # 6

1. $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n^2 + n} - 1}{n} \right)^{2\sqrt{n^2 + n} - 1}$
2. $\lim_{x \rightarrow \infty} \left(\frac{\frac{1}{a_1^x} + \frac{1}{a_2^x} + \frac{1}{a_3^x} + \dots + \frac{1}{a_n^x}}{n} \right)^{nx}$, where $a_1, a_2, \dots, a_n > 0$
3. $\lim_{x \rightarrow 0} \left[\frac{(1+x)^{1/x}}{e} \right]^{1/x}$
4. If $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p + q)$.
5. $\lim_{x \rightarrow 0} \left[\frac{\ln(1+x)^{1+x}}{x^2} - \frac{1}{x} \right]$
6. Let $L = \prod_{n=3}^{\infty} \left(1 - \frac{4}{n^2} \right)$; $M = \prod_{n=2}^{\infty} \left(\frac{n^3 - 1}{n^3 + 1} \right)$ and $N = \prod_{n=1}^{\infty} \frac{(1 + n^{-1})^2}{1 + 2n^{-1}}$, then find the value of $L^{-1} + M^{-1} + N^{-1}$.
7. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A and B. Let $T(x)$ be the area of the triangle ABC and let $S(x)$ be the area of the shaded region. Compute :



 - (a) $T(x)$
 - (b) $S(x)$ and
 - (c) the limit of $\frac{T(x)}{S(x)}$ as $x \rightarrow 0$.
8. Let $f(x) = \lim_{n \rightarrow \infty} \sum_{n=1}^n 3^{n-1} \sin^3 \frac{x}{3^n}$ and $g(x) = x - 4 f(x)$. Evaluate $\lim_{x \rightarrow 0} (1 + g(x))^{\cot x}$.

9. If $f(n, \theta) = \prod_{r=1}^n \left(1 - \tan^2 \frac{\theta}{2^r}\right)$, then compute $\lim_{n \rightarrow \infty} f(n, \theta)$
10. Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x}{e} - x \left(\frac{x}{x+1} \right)^x \right)$
11. $f(x)$ is the function such that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$. $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$, then find the value of a and b .
12. Through a point A on a circle, a chord AP is drawn and on the tangent at A a point T is taken such that $AT = AP$. If TP produced meet the diameter through A at Q , prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.
13. At the end points A, B of the fixed segment of length L , lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD . Find the value of x as θ tends to zero i.e. $\lim_{\theta \rightarrow 0} x$.
14. Let $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} \sin \frac{1}{x} + x}{1 + x^{2n}}$, then find
 (A) $\lim_{x \rightarrow \infty} x f(x)$, (B) $\lim_{x \rightarrow 1} f(x)$, (C) $\lim_{x \rightarrow 0} f(x)$, (D) $\lim_{x \rightarrow -\infty} f(x)$
15. Using Sandwich theorem, evaluate
 (a) $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$
 (b) $\lim_{n \rightarrow \infty} \frac{1}{1+n^2} + \frac{2}{2+n^2} + \dots + \frac{n}{n+n^2}$

CONTINUITY

EXERCISE # 1

[SINGLE CORRECT CHOICE TYPE]

- Let $f(x) = \begin{cases} ax+1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2+1 & \text{if } x > 1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a - b)$ is equal to-
(A) 0 (B) 1 (C) 2 (D) 4
- For the function $f(x) = \frac{1}{x + 2^{\left(\frac{1}{x-2}\right)}}$, $x \neq 2$ which of the following holds ?
(A) $f(2) = 1/2$ and f is continuous at $x = 2$ (B) $f(2) \neq 0, 1/2$ and f is continuous at $x = 2$
(C) f can not be continuous at $x = 2$ (D) $f(2) = 0$ and f is continuous at $x = 2$.
- The function $f(x) = \frac{4-x^2}{4x-x^3}$, is-
(A) Discontinuous at only one point in its domain.
(B) Discontinuous at two points in its domain.
(C) Discontinuous at three points in its domain.
(D) Continuous everywhere in its domain.
- If $f(x) = \frac{x^2 - bx + 25}{x^2 - 7x + 10}$ for $x \neq 5$ and f is continuous at $x = 5$, then $f(5)$ has the value equal to-
(A) 0 (B) 5 (C) 10 (D) 25
- If $f(x) = \frac{x - e^x + \cos 2x}{x^2}$, $x \neq 0$ is continuous at $x = 0$, then -
(A) $f(0) = \frac{5}{2}$ (B) $[f(0)] = -2$ (C) $\{f(0)\} = -0.5$ (D) $[f(0)]. \{f(0)\} = -1.5$
where $[.]$ and $\{.\}$ denotes greatest integer and fractional part function
- $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$. Then
 $\lim_{x \rightarrow a} \frac{\log_e(1 + 3f(x))}{2f(x)}$ is-
(A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$
- In $[1, 3]$, the function $[x^2 + 1]$, $[.]$ denoting the greatest integer function, is continuous -
(A) For all x
(B) For all x except at nine points
(C) For all x except at seven points
(D) For all x except at eight points

8. Number of points of discontinuity of $f(x) = [2x^3 - 5]$ in $[1, 2]$, is equal to-
(Where $[x]$ denotes greatest integer less than or equal to x)
(A) 14 (B) 13
(C) 10 (D) 8
9. Given $f(x) = \begin{cases} |x+1| & \text{if } x < -2 \\ 2x+3 & \text{if } -2 \leq x < 0 \\ x^2+3 & \text{if } 0 \leq x < 3 \\ x^3-15 & \text{if } x \geq 3 \end{cases}$. Then number of point(s) of discontinuity of $f(x)$ is-
(A) 0 (B) 1 (C) 2 (D) 3
10. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then the value of $\lim_{x \rightarrow 0} f\left(\frac{1-\cos 3x}{x^2}\right)$ is-
(A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) data insufficient
11. f is a continuous function on the real line. Given that $x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$.
Then the value of $f(\sqrt{3})$
(A) cannot be determined (B) is $2(1 - \sqrt{3})$
(C) is zero (D) is $\frac{2(\sqrt{3}-2)}{\sqrt{3}}$
12. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :
(A) all integers (B) all integers except 0 & 1
(C) all integers except 0 (D) all integers except 1
13. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function $\forall x \in \mathbb{R}$ and $f(x) = 5 \forall x \in \text{irrational}$. Then the value of $f(3)$ is -
(A) 1 (B) 2 (C) 5 (D) cannot determine
14. If $f(x) = \frac{1}{(x-1)(x-2)}$ and $g(x) = \frac{1}{x^2}$, then points of discontinuity of $f\{g(x)\}$ are -
(A) $\left\{-1, 0, 1, \frac{1}{\sqrt{2}}\right\}$ (B) $\left\{-\frac{1}{\sqrt{2}}, -1, 0, 1, \frac{1}{\sqrt{2}}\right\}$
(C) $\{0, 1\}$ (D) $\left\{0, 1, \frac{1}{\sqrt{2}}\right\}$

15. Consider the function $f(x) = \begin{cases} \frac{x}{[x]} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x = 2 \\ \sqrt{6-x} & \text{if } 2 < x \leq 3 \end{cases}$

Where $[x]$ denotes step up function then at $x = 2$ function -

- (A) has missing point removable discontinuity
- (B) has isolated point removable discontinuity
- (C) has non removable discontinuity finite type
- (D) is continuous

16. Consider $f(x) = \begin{cases} x[x]^2 \log_{(1+x)} 2, & \text{for } -1 < x < 0 \\ \frac{\ln(e^{x^2} + 2\sqrt{\{x\}})}{\tan \sqrt{x}}, & \text{for } 0 < x < 1 \end{cases}$

where $[*]$ & $\{*\}$ are the greatest integer function & fractional part function respectively, then :-

- (A) $f(0) = \ln 2 \Rightarrow f$ is continuous at $x = 0$
- (B) $f(0) = 2 \Rightarrow f$ is continuous at $x = 0$
- (C) $f(0) = e^2 \Rightarrow f$ is continuous at $x = 0$
- (D) f has an irremovable discontinuity at $x = 0$

17. The function $f(x) = [x] \cdot \cos \frac{2x-1}{2} \pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :-

(A) all x (B) all integer points (C) no x (D) x which is not an integer

18. Consider the function defined on $[0, 1] \rightarrow \mathbb{R}$, $f(x) = \frac{\sin x - x \cos x}{x^2}$ if $x \neq 0$ and $f(0) = 0$, then the function $f(x)$:-

- (A) has a removable discontinuity at $x = 0$
- (B) has a non removable finite discontinuity at $x = 0$
- (C) has a non removable infinite discontinuity at $x = 0$
- (D) is continuous at $x = 0$

19. Let $f(x) = \frac{\sqrt{x^2 + px + 1}}{x^2 - p}$. If $f(x)$ is discontinuous at exactly 2 values of x then number of integers in the range of p is

(A) 1 (B) 2 (C) 3 (D) 4

20. Let $f(x) = \begin{cases} \frac{(1 + \tan x)^{\frac{1}{x}} - e}{x}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$

If $f(x)$ is continuous at $x = 0$, then the value of k is

- (A) e (B) $\frac{-e}{2}$ (C) $\frac{-e}{4}$ (D) None

21. Let $f(x) = \begin{cases} px^2 - px + q, & x < 1 \\ x - 1, & 1 \leq x \leq 3 \\ lx^2 + mx + 2, & x > 3 \end{cases}$

If $f(x)$ is continuous $\forall x \in \mathbb{R}$, then the value of $\frac{ql - m}{l}$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

22. Let $f(x) = \begin{cases} \frac{\sin \alpha x^2}{x^2}, & x \neq 0 \\ \frac{3}{4} + \frac{1}{4\alpha}, & x = 0 \end{cases}$

If $f(x)$ is continuous at $x = 0$, then α can be

- (A) 1 (B) 2 (C) 3 (D) 4

23. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function and assumes only rational values. If $f(0) = 2$ then

the value of $\tan^{-1}\left(f\left(\frac{1}{2}\right)\right) + \tan^{-1}\left(\frac{3}{2}f\left(\frac{1}{2}\right)\right)$ is

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) π

24. $f(x) = \begin{cases} x + 1; & x < 0 \\ \cos x; & x \geq 0 \end{cases}$ at $x = 0$, $f(x)$ is

- (A) continuous (B) having removable discontinuity
(C) discontinuous (D) none

EXERCISE # 2

1. If the function $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ is continuous at $x = -2$. Find $f(-2)$.
2. Find all possible values of a and b so that $f(x)$ is continuous for all $x \in \mathbb{R}$ if

$$f(x) = \begin{cases} |ax+3| & \text{if } x \leq -1 \\ |3x+a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases}$$

$$3. \quad \text{The function } f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b+2 & \text{if } x = \frac{\pi}{2} \\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$$

Determine the values of ' a ' & ' b ', if f is continuous at $x = \pi/2$.

$$4. \quad \text{Suppose that } f(x) = x^3 - 3x^2 - 4x + 12 \text{ and } h(x) = \begin{cases} \frac{f(x)}{x-3} & , \quad x \neq 3 \\ K & , \quad x = 3 \end{cases} \text{ then}$$

- (a) find all zeroes of $f(x)$.
- (b) find the value of K that makes h continuous at $x = 3$.
- (c) using the value of K found in (b) determine whether h is an even function.

$$5. \quad \text{Let } f(x) = \begin{cases} \frac{1 - \sin \pi x}{1 + \cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases} . \text{ Determine the value of } p, \text{ if possible, so that the}$$

function is continuous at $x = 1/2$.

6. Given the function $g(x) = \sqrt{6-2x}$ and $h(x) = 2x^2 - 3x + a$. Then
 (a) evaluate $h(g(2))$ (b) If $f(x) = \begin{cases} g(x), & x \leq 1 \\ h(x), & x > 1 \end{cases}$, find 'a' so that f is continuous.
7. Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3 \end{cases}$. Determine the form of $g(x) = f[f(x)]$ & hence find the point of discontinuity of g , if any.
8. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2} - 1} & \text{if } x > 0 \\ \frac{e^{\sin 4x} - 1}{\ln(1 + \tan 2x)} & \text{if } x < 0 \end{cases}$
 Is it possible to define $f(0)$ to make the function continuous at $x = 0$. If yes what is the value of $f(0)$, if not then indicate the nature of discontinuity.
9. Determine a & b so that f is continuous at $x = \frac{\pi}{2}$ where $f(x) = \begin{cases} \frac{1 - \sin^3 x}{3 \cos^2 x} & \text{if } x < \frac{\pi}{2} \\ a & \text{if } x = \frac{\pi}{2} \\ \frac{b(1 - \sin x)}{(\pi - 2x)^2} & \text{if } x > \frac{\pi}{2} \end{cases}$
10. Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x + bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$
 is continuous at $x = 0$.

EXERCISE # 3 (JM)

1. The function $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$ can be made continuous at $x = 0$ by defining $f(0)$ as- [AIEEE 2007]
 (1) 2 (2) -1 (3) 0 (4) 1

2. The values of p and q for which the function $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{\frac{3}{2}}}, & x > 0 \end{cases}$

is continuous for all x in $\mathbb{R}$ are :-

[AIEEE 2011]

- (1) $p = -\frac{3}{2}, q = \frac{1}{2}$ (2) $p = \frac{1}{2}, q = \frac{3}{2}$ (3) $p = \frac{1}{2}, q = -\frac{3}{2}$ (4) $p = \frac{5}{2}, q = \frac{1}{2}$

3. Define $F(x)$ as the product of two real functions $f_1(x) = x, x \in \mathbb{R}$, and $f_2(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ as follows : $F(x) = \begin{cases} f_1(x) \cdot f_2(x) & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ [AIEEE 2011]

Statement-1 : $F(x)$ is continuous on $\mathbb{R}$.

Statement-2 : $f_1(x)$ and $f_2(x)$ are continuous on $\mathbb{R}$.

- (1) Statement-1 is false, statement-2 is true.
 (2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement1.
 (3) Statement-1 is true, statement-2 is true, statement-2 is not a correct explanation for statement1
 (4) Statement-1 is true, statement-2 is false

4. If $f(x)$ is continuous and $f(9/2) = 2/9$, then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to : [JEE Mains Offline-2014]
 (1) 9/2 (2) 0 (3) 2/9 (4) 8/9

5. If the function $f(x) = \begin{cases} \frac{\sqrt{2 + \cos x} - 1}{(\pi - x)^2}, & x \neq \pi \\ k, & x = \pi \end{cases}$ is continuous at $x = \pi$, then k equals:-
 (1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 2 (4) 0

[JEE Mains Offline-2014]

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as :

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

Then, f is :

(1) continuous if $a = 5$ and $b = 5$

(2) continuous if $a = -5$ and $b = 10$

(3) continuous if $a = 0$ and $b = 5$

(4) not continuous for any values of a and b

[JEE Mains Online-2019]

7. Let $f : [-1, 3] \rightarrow \mathbb{R}$ be defined as

[JEE Mains Online-2019]

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3, \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then, f is discontinuous at :

(1) only two points (2) only three points (3) only one point (4) four or more points

8. If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by $f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$ is continuous, then k is

equal to :

[JEE Mains Online-2019]

(1) $\frac{1}{\sqrt{2}}$

(2) $\frac{1}{2}$

(3) 1

(4) 2

9. If $f(x) = [x] - \left[\frac{x}{4}\right]$, $x \in \mathbb{R}$ where $[x]$ denotes the greatest integer function, then :

[JEE Mains Online-2019]

(1) $\lim_{x \rightarrow 4^-} f(x)$ exists but $\lim_{x \rightarrow 4^+} f(x)$ does not exist (2) $\lim_{x \rightarrow 4^-} f(x)$ exists but $\lim_{x \rightarrow 4^+} f(x)$ does not exist.

(3) f is continuous at $x = 4$.

(4) Both $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$ exist but are not equal.

10. If the function $f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$

Is continuous at $x = 5$, then the value of $a - b$ is :

[JEE Mains Online-2019]

(1) $\frac{2}{\pi - 5}$

(2) $\frac{2}{\pi + 5}$

(3) $\frac{-2}{\pi + 5}$

(4) $\frac{2}{5 - \pi}$

11. If $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$ is continuous at $x = 0$, then the

ordered pair (p, q) is equal to :

[JEE Mains Online-2019]

(1) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$

(2) $\left(\frac{5}{2}, \frac{1}{2}\right)$

(3) $\left(-\frac{3}{2}, \frac{1}{2}\right)$

(4) $\left(-\frac{1}{2}, \frac{3}{2}\right)$

12. If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by

$$f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+3x}{1-2x} \right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$$

is continuous, then k is equal to

[JEE Mains Online-2020]

13. If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} & ; x < 0 \\ b & ; x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}} & ; x > 0 \end{cases}$

is continuous at $x = 0$, find $a + 2b$ is equal to :

(1) -2

(2) 0

(3) -1

[JEE Mains Online-2020]

(4) 1

14. Let $[t]$ denote the greatest integer $\leq t$ and $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$. Then the function, $f(x) = [x^2] \sin(\pi x)$ is discontinuous, when x is equal to :

[JEE Mains Online-2020]

(1) $\sqrt{A}$

(2) $\sqrt{A+5}$

(3) $\sqrt{A+21}$

(4) $\sqrt{A+1}$

EXERCISE # 4 (JA)

SECTION-1

1. For every integer n , let a_n and b_n be real numbers. Let function $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}, \text{ for all integers } n.$$

If f is continuous, then which of the following holds(s) for all n ?

[JEE 2012, 4M]

- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$
(C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

2. For every pair of continuous function $f, g : [0, 1] \rightarrow \mathbb{R}$ such that

$$\max\{f(x) : x \in [0, 1]\} = \max\{g(x) : x \in [0, 1]\},$$

the correct statement(s) is(are) :

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
(D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

[JEE(Advanced)-2014, 3]

3. Let $[x]$ be the greatest integer less than or equals to x . Then, at which of the following point(s)

the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous ?

[JEE(Advanced)-2017]

- (A) $x = -1$ (B) $x = 0$
(C) $x = 2$ (D) $x = 1$

SECTION-2

4. Discuss the continuity of the function $f(x) = \begin{cases} \frac{e^{1/(x-1)} - 2}{e^{1/(x-1)} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases}$ at $x = 1$.

[REE 2001 (Mains), 3 out of 100]

EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- Which of the following function(s) is/are discontinuous at $x = 0$?
 (A) $f(x) = \sin \frac{\pi}{2x}$, $x \neq 0$ and $f(0) = 1$ (B) $g(x) = x \sin \left(\frac{\pi}{x} \right)$, $x \neq 0$ and $g(0) = \pi$
 (C) $h(x) = \frac{|x|}{x}$, $x \neq 0$ and $h(0) = 1$ (D) $k(x) = \frac{1}{1+e^{\cot x}}$, $x \neq 0$ and $k(0) = 0$.
- A function $f(x)$ is defined as $f(x) = \frac{A \sin x + \sin 2x}{x^3}$, ($x \neq 0$). If the function is continuous at $x = 0$, then -
 (A) $A = -2$ (B) $f(0) = -1$ (C) $A = 1$ (D) $f(0) = 1$
- Which of the following function(s) not defined at $x = 0$ has/have non-removable discontinuity at the point $x = 0$?
 (A) $f(x) = \frac{1}{1+2^{\frac{1}{x}}}$ (B) $f(x) = \arctan \frac{1}{x}$
 (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{\frac{1}{e^x} + 1}$ (D) $f(x) = \frac{1}{\ln |x|}$
- Which of the following function(s) not defined at $x = 0$ has/have removable discontinuity at $x = 0$?
 (A) $f(x) = \frac{1}{1+2^{\cot x}}$ (B) $f(x) = \cos \left(\frac{|\sin x|}{x} \right)$ (C) $f(x) = x \sin \frac{\pi}{x}$ (D) $f(x) = \frac{1}{\ln |x|}$
- Let $f(x) = \begin{cases} \frac{e^x - 1 + ax}{x^2}, & x > 0 \\ b, & x = 0, \text{ then -} \\ \frac{\sin \frac{x}{2}}{x}, & x < 0 \end{cases}$
 (A) $f(x)$ is continuous at $x = 0$ if $a = -1$, $b = \frac{1}{2}$.
 (B) $f(x)$ is discontinuous at $x = 0$, if $b \neq \frac{1}{2}$.
 (C) $f(x)$ has non-removable discontinuity at $x = 0$ if $a \neq -1$.
 (D) $f(x)$ has removable discontinuity at $x = 0$ if $a = -1$, $b \neq \frac{1}{2}$

6. Which of the following function(s) has removable type of discontinuity at $x = 0$?
- (A) $f(x) = \frac{1 - \sec^2 2x}{4x^2}$ (B) $g(x) = \frac{\csc x - 1}{x \csc x}$ (where $\csc x = \operatorname{cosec} x$)
- (C) $h(x) = \frac{\sin 5x}{x}$ (D) $\ell(x) = (1 + 2x^2)^{\frac{1}{x^2}}$
7. If f is defined on an interval $[a, b]$. Which of the following statement(s) is/are INCORRECT ?
- (A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (B) If f is continuous on $[a, b]$, $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (C) If f is continuous on $[a, b]$ and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.
 (D) If f has no zeroes on $[a, b]$, then $f(a)$ and $f(b)$ have the same sign.
8. Which of the following functions f has/have a removable discontinuity at the indicated point ?
- (A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at $x = -2$ (B) $f(x) = \frac{x - 7}{|x - 7|}$ at $x = 7$
- (C) $f(x) = \frac{x^3 + 64}{x + 4}$ at $x = -4$ (D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at $x = 9$
9. In which of the following cases the given equations has atleast one root in the indicated interval ?
- (A) $x - \cos x = 0$ in $(0, \pi/2)$
 (B) $x + \sin x = 1$ in $(0, \pi/6)$
 (C) $\frac{a}{x-1} + \frac{b}{x-3} = 0$, $a, b > 0$ in $(1, 3)$
 (D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$.
10. Indicate all correct alternatives if, $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$
- (A) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both continuous
 (B) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both discontinuous
 (C) $\tan(f(x))$ & $f^{-1}(x)$ are both continuous
 (D) $\tan(f(x))$ is continuous but $\frac{1}{f(x)}$ is not

[MATRIX TYPE]

11. Column-I

Column-II

(A) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\ln x}$ is

(P) 2

(B) $\lim_{x \rightarrow 0} \frac{x(\cos x - \cos 2x)}{2 \sin x - \sin 2x}$ is

(Q) 3

(C) $\lim_{x \rightarrow 0} \frac{\tan x \sqrt{\tan x} - \sin x \sqrt{\sin x}}{x^3 \sqrt{x}}$ is

(R) $\frac{3}{2}$

(D) If $f(x) = \cos \left(x \cos \frac{1}{x}\right)$ and $g(x) = \frac{\ln(\sec^2 x)}{x \sin x}$

(S) $\frac{3}{4}$

are both continuous at $x = 0$ then $f(0) + g(0)$ equals

12. Match the function in **Column-I** with its behaviour at $x = 0$ in **column-II**, where $[.]$ denotes greatest integer function & $\text{sgn}(x)$ denotes signum function.

Column-I

Column-II

(A) $f(x) = [x][1 + x]$

(P) LHL exist at $x = 0$

(B) $f(x) = [-x][1 + x]$

(Q) RHL exist at $x = 0$

(C) $f(x) = (\text{sgn}(x))[2 - x][1 + |x|]$

(R) Continuous at $x = 0$

(D) $f(x) = [\cos x]$

(S) $\lim_{x \rightarrow 0} f(x)$ exists but function is discontinuous at

$x = 0$

(T) $\lim_{x \rightarrow 0} f(x)$ does not exist

EXERCISE # 6

1. If $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) is cont. at $x = 0$. Find A and B. Also find $f(0)$.

2. Let $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{x\}^2)\right) \sin^{-1}(1 - \{x\})}{\sqrt{2}(\{x\} - \{x\}^3)} & \text{for } x \neq 0 \\ \frac{\pi}{2} & \text{for } x = 0 \end{cases}$ where $\{x\}$ is the fractional part of x .

Consider another function $g(x)$; such that

$$g(x) = \begin{cases} f(x) & \text{for } x \geq 0 \\ 2\sqrt{2}f(x) & \text{for } x < 0 \end{cases} \quad \text{Discuss the continuity of the functions } f(x) \text{ \& } g(x) \text{ at } x = 0.$$

3. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x . Discuss the continuity of f in $[-2, 2]$.

4. Find the locus of (a, b) for which the function $f(x) = \begin{cases} ax - b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2 - a & \text{for } x \geq 2 \end{cases}$

is continuous at $x = 1$ but discontinuous at $x = 2$.

5. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ where f is continuous on $\mathbb{R}$.

Find the values of a , b and c .

6. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\operatorname{cosec} x} & ; \quad -\frac{\pi}{2} < x < 0 \\ a & ; \quad x = 0 \\ \frac{e^{1/x} + e^{2/x} + e^{3/|x|}}{ae^{2/x} + be^{3/|x|}} & ; \quad 0 < x < \frac{\pi}{2} \end{cases}$

If $f(x)$ is continuous at $x = 0$, find the value of $(a^2 + b^2)$.

7. Given $f(x) = \sum_{r=1}^n \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right); r, n \in \mathbb{N}$

$$g(x) = \lim_{n \rightarrow \infty} \frac{\ln\left(f(x) + \tan \frac{x}{2^n}\right) - \left(f(x) + \tan \frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan \frac{x}{2}\right)\right]}{1 + \left(f(x) + \tan \frac{x}{2^n}\right)^n}$$

$$= k \text{ for } x = \frac{\pi}{4} \text{ and the domain of } g(x) \text{ is } (0, \pi/2).$$

where $[]$ denotes the greatest integer function.

Find the value of k , if possible, so that $g(x)$ is continuous at $x = \pi/4$. Also state the points of discontinuity of $g(x)$ in $(0, \pi/4)$, if any.

8. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \frac{f(x)}{g(x)}$, where h is a rational function such that

(a) it is continuous every where except when $x = -1$,

(b) $\lim_{x \rightarrow \infty} h(x) = \infty$ and

(c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

Find $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x))$

9. (a) Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \forall x \in \mathbb{R}$.

If $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$.

(b) Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$ where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively.

10. (a) If $g : [a, b] \rightarrow [a, b]$ is continuous & onto function, then show that there is some $c \in [a, b]$ such that $g(c) = c$.

(b) Let f be continuous on the interval $[0, 1]$ to $\mathbb{R}$ such that $f(0) = f(1)$. Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$

DIFFERENTIABILITY

EXERCISE # 1

[SINGLE CORRECT CHOICE TYPE]

1. Let $f(x) = [\tan^2 x]$, (where $[.]$ denotes greatest integer function). Then -
 (A) $\lim_{x \rightarrow 0} f(x)$ does not exist (B) $f(x)$ is continuous at $x = 0$.
 (C) $f(x)$ is not differentiable at $x = 0$ (D) $f'(0) = 1$
2. The number of points where $f(x) = [\sin x + \cos x]$ (where $[.]$ denotes the greatest integer function), $x \in (0, 2\pi)$ is not continuous is -
 (A) 3 (B) 4 (C) 5 (D) 6
3. If 6, 8 and 12 are ℓ^{th} , m^{th} and n^{th} terms of an A.P. and $f(x) = nx^2 + 2\ell x - 2m$, then the equation $f(x) = 0$ has -
 (A) a root between 0 and 1 (B) both roots imaginary.
 (C) both roots negative. (D) both roots greater than 1.
4. Let f be differentiable at $x = 0$ and $f'(0) = 1$. Then $\lim_{h \rightarrow 0} \frac{f(h) - f(-2h)}{h} =$
 (A) 3 (B) 2 (C) 1 (D) -1
5. Let $g(x) = \begin{cases} 3x^2 - 4\sqrt{x} + 1 & \text{for } x < 1 \\ ax + b & \text{for } x \geq 1 \end{cases}$
 If $g(x)$ is continuous and differentiable for all numbers in its domain then -
 (A) $a = b = 4$ (B) $a = b = -4$ (C) $a = 4$ and $b = -4$ (D) $a = -4$ and $b = 4$
6. If $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ and $f(1) = 2$, $f'(1) = 2$ then $\text{sgn } f(x)$ is equal to (where sgn denotes signum function) -
 (A) 0 (B) 1 (C) -1 (D) 4
7. The function $g(x) = \begin{cases} x + b, & x < 0 \\ \cos x & x \geq 0 \end{cases}$ can be made differentiable at $x = 0$ -
 (A) if b is equal to zero (B) if b is not equal to zero
 (C) if b takes any real value (D) for no value of b
8. Which one of the following functions is continuous everywhere in its domain but has at least one point where it is not differentiable ?
 (A) $f(x) = x^{1/3}$ (B) $f(x) = \frac{|x|}{x}$ (C) $f(x) = e^{-x}$ (D) $f(x) = \tan x$
9. If the right hand derivative of $f(x) = [x] \tan \pi x$ at $x = 7$ is $k\pi$, then k is equal to ($[y]$ denotes greatest integer $\leq y$)
 (A) 6 (B) 7 (C) -7 (D) 49

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous onto function satisfying $f(x) + f(-x) = 0, \forall x \in \mathbb{R}$. If $f(-3) = 2$ and $f(5) = 4$ in $[-5, 5]$, then the equation $f(x) = 0$ has-
 (A) exactly three real roots (B) exactly two real roots
 (C) atleast five real roots (D) atleast three real roots

11. Let $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{ax(x-1)\left(\cot \frac{\pi x}{4}\right)^n + (px^2 + 2)}{\left(\cot \frac{\pi x}{4}\right)^n + 1}, & x \in (0, 1) \cup (1, 2) \\ 0, & x = 1 \end{cases}$

If $f(x)$ is differentiable for all $x \in (0, 2)$ then $(a^2 + p^2)$ equals -

- (A) 18 (B) 20 (C) 22 (D) 24
12. If $2x + 3|y| = 4y$, then y as a function of x i.e. $y = f(x)$, is -
 (A) discontinuous at one point
 (B) non differentiable at one point
 (C) discontinuous & non differentiable at same point
 (D) continuous & differentiable everywhere

13. If $f(x) = (x^5 + 1)|x^2 - 4x - 5| + \sin|x| + \cos(|x - 1|)$, then $f(x)$ is not differentiable at -
 (A) 2 points (B) 3 points (C) 4 points (D) zero points

14. Let $f(x) = \begin{cases} x^3 + 2x^2 & x \in \mathbb{Q} \\ -x^3 + 2x^2 + ax & x \notin \mathbb{Q} \end{cases}$, then the integral value of 'a' so that $f(x)$ is differentiable at $x = 1$, is
 (A) 2 (B) 6 (C) 7 (D) not possible

15. Let $\mathbb{R}$ be the set of real numbers and $f: \mathbb{R} \rightarrow \mathbb{R}$, be a differentiable function such that $|f(x) - f(y)| \leq |x - y|^3 \forall x, y \in \mathbb{R}$. If $f(10) = 100$, then the value of $f(20)$ is equal to -
 (A) 0 (B) 10 (C) 20 (D) 100

16. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases} \text{ is differentiable for all real } x ?$$

- (A) $\{(a, 1 - 2a, a) \mid a \in \mathbb{R}, a \neq 0\}$
 (B) $\{(a, 1 - 2a, c) \mid a, c \in \mathbb{R}, a \neq 0\}$
 (C) $\{(a, b, c) \mid a, b, c \in \mathbb{R}, a + b + c = 1\}$
 (D) $\{(a, 1 - 2a, 0) \mid a \in \mathbb{R}, a \neq 0\}$
17. Number of points of non-differentiability of the function
 $g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ in $(-50, 50)$ where $[x]$ and $\{x\}$ denotes the greatest integer function and fractional part function of x respectively, is equal to:-
 (A) 98 (B) 99 (C) 100 (D) 0

18. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in \mathbb{I}$ and p is a prime number. The number of points where $f(x)$ is not differentiable is :-
 (A) $p - 1$ (B) $p + 1$ (C) $2p + 1$ (D) $2p - 1$
 Here $[x]$ denotes greatest integer function.
19. The function $f(x) = (x^2 - 1) |x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at :
 (A) -1 (B) 0 (C) 1 (D) 2
20. Let $g(x) = \begin{cases} 2x + \tan^{-1} x + a, & -\infty < x \leq 0 \\ x^3 + x^2 + bx, & 0 < x < \infty \end{cases}$.
 If $g(x)$ is differentiable for all $x \in (-\infty, \infty)$ then $(a^2 + b^2)$ is equal to
 (A) 20 (B) 13 (C) 9 (D) 4
21. If the function $f(x) = \begin{cases} ax + b, & -\infty < x \leq 2 \\ x^2 - 5x + 6, & 2 < x < 3 \\ px^2 + qx + 1, & 3 \leq x < \infty \end{cases}$
 is differentiable in $(-\infty, \infty)$, then
 (A) $a = -1, p = \frac{-4}{9}$ (B) $b = 2, q = \frac{5}{3}$ (C) $a = 1, b = 2$ (D) $a = -1, q = \frac{-5}{3}$
22. Let $f(x) = [x]$ and $g(x) = \begin{cases} x, & x \in [0, 1) \\ x - 1, & x \in [1, 2) \\ x - 2, & x \in [2, 3) \\ 0, & x = 3 \end{cases}$.
 Then $f(x) + g(x)$ is
 (A) discontinuous at $x = 1$ and $x = 2$.
 (B) continuous in $[0, 3]$ but non derivable in $[0, 3]$.
 (C) not twice differentiable in $[0, 3]$.
 (D) twice differentiable in $[0, 3]$
 [Note : $[k]$ denotes the greatest integer function less than or equal to k .]
23. Let $f(x) = \begin{cases} x + 2, & x < 0 \\ -(2 + x^2), & 0 \leq x < 1 \\ x, & x \geq 1 \end{cases}$
 Then the number of points where $|f(x)|$ is non-derivable is
 (A) 3 (B) 2 (C) 1 (D) 0
24. Let $g(x) = \min(x, x^2)$ where $x \in \mathbb{R}$, then $\lim_{x \rightarrow 0} \frac{g(1+x) - g(1)}{x}$ equals
 (A) 0 (B) 1 (C) 2 (D) does not exist

EXERCISE # 2

- Discuss the continuity & differentiability of the function $f(x) = \sin x + \sin |x|$, $x \in \mathbb{R}$. Draw a rough sketch of the graph of $f(x)$.
- Examine the continuity and differentiability of $f(x) = |x| + |x-1| + |x-2|$, $x \in \mathbb{R}$. Also draw the graph of $f(x)$.

- If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then find the range of n .

- A function f is defined as follows: $f(x) = \begin{cases} 1 & \text{for } -\infty < x < 0 \\ 1 + |\sin x| & \text{for } 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & \text{for } \frac{\pi}{2} \leq x < +\infty \end{cases}$

Discuss the continuity & differentiability at $x = 0$ & $x = \pi/2$.

- Examine the origin for continuity & derivability in the case of the function f defined by $f(x) = x \tan^{-1}(1/x)$, $x \neq 0$ and $f(0) = 0$.
- Let $f(0) = 0$ and $f'(0) = 1$. For a positive integer k , show that $\lim_{x \rightarrow 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + \dots + f\left(\frac{x}{k}\right) \right) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$
- Let $f(x) = x e^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}$; $x \neq 0$, $f(0) = 0$, test the continuity & differentiability at $x = 0$
- If $f(x) = |x-1| \cdot ([x] - [-x])$, then find $f'(1^+)$ & $f'(1^-)$ where $[x]$ denotes greatest integer function.

- If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at $x = 1$. Find the values of a & b .

- Let $g(x) = \begin{cases} a\sqrt{x+2}, & 0 < x < 2 \\ bx+2, & 2 \leq x < 5 \end{cases}$. If $g(x)$ is derivable on $(0, 5)$, then find $(2a + b)$.

EXERCISE # 3 (JM)

1. The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable- [AIEEE-2006]
 - (1) $(-\infty, -1) \cup (-1, \infty)$
 - (2) $(-\infty, \infty)$
 - (3) $(0, \infty)$
 - (4) $(-\infty, 0) \cup (0, \infty)$

2. Let $f(x) = x|x|$ and $g(x) = \sin x$. [AIEEE-2009]

Statement-1 : $g \circ f$ is differentiable at $x = 0$ and its derivative is continuous at that point.

Statement-2 : $g \circ f$ is twice differentiable at $x = 0$.

 - (1) Statement – 1 is true, Statement – 2 is false.
 - (2) Statement – 1 is false, Statement – 2 is true.
 - (3) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1.
 - (4) Statement – 1 is true, Statement – 2 is true ; Statement – 2 is not a correct explanation for statement – 1.

3. If function $f(x)$ is differentiable at $x = a$ then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ [AIEEE-2011]
 - (1) $2a f(a) + a^2 f'(a)$
 - (2) $-a^2 f'(a)$
 - (3) $a f(a) - a^2 f'(a)$
 - (4) $2af(a) - a^2 f'(a)$

4. Consider the function,
 $f(x) = |x - 2| + |x - 5|, x \in \mathbb{R}$.
Statement – 1 : $f'(4) = 0$.
Statement – 2 : f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$. [AIEEE 2012]
 - (1) Statement-1 is true, Statement-2 is false.
 - (2) Statement-1 is false, Statement-2 is true.
 - (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement1.
 - (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement1.

5. Let $f(x) = x|x|$, $g(x) = \sin x$ and $h(x) = (g \circ f)(x)$. Then

[On-line 2014]

- (1) $h'(x)$ is differentiable at $x = 0$
- (2) $h'(x)$ is continuous at $x = 0$ but is not differentiable at $x = 0$
- (3) $h(x)$ is differentiable at $x = 0$ but $h'(x)$ is not continuous at $x = 0$
- (4) $h(x)$ is not differentiable at $x = 0$

6. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, and $g(x) = x f(x) :-$

Statement I : f is a continuous function at $x = 0$.

[On-line 2014]

Statement II : g is a differentiable function at $x = 0$.

- (1) Statement I is false and statement II is true
- (2) Statement I is true and statement II is false
- (3) Both statement I and II are true
- (4) Both statements I and II are false

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq x^2$, for all $x \in \mathbb{R}$. Then, at $x = 0$, f is:

- (1) Neither continuous nor differentiable
- (2) Differentiable but not continuous
- (3) Continuous as well as differentiable
- (4) Continuous but not differentiable

[On-line 2014]

8. Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x) - f(y)| \leq 2|x - y|^{3/2}$, for all

$x, y \in \mathbb{R}$. If $f(0) = 1$ then $\int_0^1 f^2(x) dx$ is equal to :

[JEE Mains Online-2019]

- (1) 1
- (2) 2
- (3) $\frac{1}{2}$
- (4) 0

9. Let $f(x) = \begin{cases} \max\{|x|, x^2\}, & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$

Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S :

[JEE Mains Online-2019]

- (1) equals $(-2, -1, 1, 2)$
- (2) equals $\{-2, 2\}$
- (3) is an empty set
- (4) equals $\{-2, -1, 0, 1, 2\}$

10. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max \{-|x|, -\sqrt{1-x^2}\}$. If K be the set of all points at which f is not differentiable, then K has exactly: **[JEE Mains Online-2019]**
 (1) one element (2) two elements (3) five elements (4) three elements
11. Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1 & 0 \leq x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is : **[JEE Mains Online-2019]**
 (1) not continuous
 (2) not differentiable at one point
 (3) differentiable at all points
 (4) not differentiable at two points
12. Let K be the set of all real values of x where the function $f(x) = \sin|x| - |x| + 2(x - \pi) \cos|x|$ is not differentiable. Then the set K is equal to: **[JEE Mains Online-2019]**
 (1) $\{\pi\}$ (2) ϕ (an empty set) (3) $\{0\}$ (4) $\{0, \pi\}$
13. Let S be the set of all points in $(-\pi, \pi)$ at which the function, $f(x) = \min \{\sin x, \cos x\}$ is not differentiable. Then S is a subset of which of the following ? **[JEE Mains Online-2019]**
 (1) $\left\{-\frac{\pi}{2}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}\right\}$ (2) $\left\{-\frac{\pi}{4}, 0, \frac{\pi}{4}\right\}$
 (3) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (4) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}$
14. Let $f(x) = 15 - |x - 10|$; $x \in \mathbb{R}$. Then the set of all values of x , at which the function, $g(x) = f(f(x))$ is not differentiable is: **[JEE Mains Online-2019]**
 (1) $\{10, 15\}$ (2) $\{10\}$ (3) $\{5, 10, 15, 20\}$ (4) $\{5, 10, 15\}$
15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is : **[JEE Mains Online-2019]**
 (1) differentiable if $f'(c) = 0$
 (2) not differentiable
 (3) not differentiable if $f'(c) = 0$
 (4) differentiable if $f'(c) \neq 0$
16. Let S be the set of points where the function, $f(x) = |2 - |x - 3||$, $x \in \mathbb{R}$, is not differentiable. Then $\sum_{x \in S} f(f(x))$ is equal to _____. **[JEE Mains Online-2020]**

EXERCISE # 4 (JA)

SECTION-1

1. Let $g(x) = \frac{(x-1)^n}{\ln \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$ and let p be the left hand derivative of $|x - 1|$ at $x = 1$. If $\lim_{x \rightarrow 1^+} g(x) = p$, then :- [JEE 2008, 3]
 (A) $n = 1, m = 1$ (B) $n = 1, m = -1$ (C) $n = 2, m = 2$ (D) $n > 2, m = n$
2. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in \mathbb{R}$, then f is - [JEE 2012, 3M, -1M]
 (A) Differentiable both at $x = 0$ and at $x = 2$
 (B) Differentiable at $x = 0$ but not differentiable at $x = 2$
 (C) Not differentiable at $x = 0$ but differentiable at $x = 2$
 (D) Differentiable neither at $x = 0$ nor at $x = 2$

SECTION-2

3. If $f(x) = \min. (1, x^2, x^3)$, then [JEE 2006, 5]
 (A) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (B) $f'(x) > 0, \forall x > 1$
 (C) $f(x)$ is not differentiable but continuous $\forall x \in \mathbb{R}$
 (D) $f(x)$ is not differentiable for two values of x
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y), \forall x, y \in \mathbb{R}$.
 If $f(x)$ is differentiable at $x = 0$, then
 (A) $f(x)$ is differentiable only in a finite interval containing zero
 (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (C) $f'(x)$ is constant $\forall x \in \mathbb{R}$
 (D) $f(x)$ is differentiable except at finitely many points [JEE 2011, 4M]
5. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x - 1, & 0 < x \leq 1 \\ \ln x, & x > 1 \end{cases}$ then - [JEE 2011, 4M]
 (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$ (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 1$ (D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

6. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases} \text{ and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0 \end{cases}$$

List-I

- P. f_4 is
Q. f_3 is
R. $f_2 \circ f_1$ is
S. f_2 is

List-II

1. onto but not one-one
2. neither continuous nor one-one
3. differentiable but not one-one
4. continuous and one-one

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

[JEE(Advanced)-2014, 3(-1)]

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define

$$h : \mathbb{R} \rightarrow \mathbb{R} \text{ by } h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

[JEE(Advanced)-2014, 3]

8. Let $a, b \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is -

- (A) Differentiable at $x = 0$ if $a = 0$ and $b = 1$
(B) Differentiable at $x = 1$ if $a = 1$ and $b = 0$
(C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
(D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

[JEE(Advanced)-2016, 4(-2)]

9. Let $f : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 3]$ and $g(x) = |x| f(x) + |4x - 7| f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then.

[JEE(Advanced)-2016, 4(-2)]

- (A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$
(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$
(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$
(D) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$. Then, the value of $\log_e(f(4))$ is _____.

[JEE Advanced-2018, 3(0)]

11. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined

by

[JEE Advanced-2018, 3(-1)]

(i) $f_1(x) = \sin\left(\sqrt{1-e^{-x^2}}\right),$

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases},$ where the inverse trigonometric function $\tan^{-1}x$ assumes

$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ values in

(iii) $f_4(x) = [\sin(\log_e(x+2))]$, where, for $t \in \mathbb{R}$, $[t]$ denotes the greatest integer less than or equal to t ,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

List-I

- P. The function f_1 is
Q. The function f_2 is
R. The function f_3 is
S. The function f_4 is

List-II

1. NOT continuous at $x = 0$
2. continuous at $x = 0$ and NOT differentiable at $x = 0$
3. differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$
4. differentiable at $x = 0$ and its derivative is continuous at $x = 0$

The correct option is :

- (A) $P \rightarrow 2 ; Q \rightarrow 3 ; R \rightarrow 1 ; S \rightarrow 4$
(C) $P \rightarrow 4 ; Q \rightarrow 2 ; R \rightarrow 1 ; S \rightarrow 3$

- (B) $P \rightarrow 4 ; Q \rightarrow 1 ; R \rightarrow 2 ; S \rightarrow 3$
(D) $P \rightarrow 2 ; Q \rightarrow 1 ; R \rightarrow 4 ; S \rightarrow 3$

EXERCISE # 5

[MULTIPLE CORRECT CHOICE TYPE]

- The function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \left(\frac{x^2}{4}\right) - \left(\frac{3x}{2}\right) + \left(\frac{13}{4}\right), & x < 1 \end{cases}$ is -

(A) continuous at $x = 1$ (B) differentiable at $x = 1$
(C) continuous at $x = 3$ (D) differentiable at $x = 3$
- Select the correct statements -

(A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.

(B) The function $f(x) = x^2|x|$ is twice differentiable at $x = 0$.

(C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim_{x \rightarrow 2} f(4x^2 - 11)$ exists

(D) If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ need not exist.
- If $f(x) = \text{sgn}(x^5)$, then which of the following is/are false (where sgn denotes signum function) -

(A) $f'(0^+) = 1$
(B) $f'(0^-) = -1$
(C) f is continuous but not differentiable at $x = 0$
(D) f is discontinuous at $x = 0$
- Graph of $f(x)$ is shown in adjacent figure, then in $[0, 5]$

(A) $f(x)$ has non removable discontinuity at two points
(B) $f(x)$ is non differentiable at four points
(C) $\lim_{x \rightarrow 1} f(f(x)) = 1$
(D) Number of points of discontinuity = number of points of non-differentiability
- Let S denotes the set of all points where $\sqrt[5]{x^2|x^3|} - \sqrt[3]{x^2|x|} - 1$ is not differentiable then S is a subset of -

(A) $\{0, 1\}$ (B) $\{0, 1, -1\}$ (C) $\{0, 1\}$ (D) $\{0\}$

6. Which of the following statements is/are correct ?
- (A) There exist a function $f : [0,1] \rightarrow \mathbb{R}$ which is discontinuous at every point in $[0,1]$ & $|f(x)|$ is continuous at every point in $[0,1]$
- (B) Let $F(x) = f(x) \cdot g(x)$. If $f(x)$ is differentiable at $x = a$, $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.
- (C) If $f'(a^+) = 2$ & $f'(a^-) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$
- (D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a, b) provided f is continuous on $[a, b]$
7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then which of the following is/are not always true-
- (A) If f is continuous then g is also continuous
- (B) If f is one-one then g is also one-one
- (C) If f is onto then g is also onto
- (D) If f is differentiable then g is also differentiable
8. If function defined by $f(x) = \begin{cases} \frac{(x-m)}{|x-m|}, & x \leq 0 \\ 2x^2 + 3ax + b, & 0 < x < 1 \\ m^2x + b - 2, & x \geq 1 \end{cases}$ is continuous & differentiable everywhere, then
- (A) $b + m = -1$ (B) $b + m = 1$ (C) $b + m = -3$ (D) $m^2 + a + b = 3$
9. The function $\phi(x) = [x] - \sin|x|$ (where $[.]$ denotes greater integer function) is -
- (A) derivable at $x = 0$
- (B) continuous at $x = 0$
- (C) $\lim_{x \rightarrow 0} \phi(x)$ does not exist
- (D) continuous and derivable at $x = 0$
10. Let $f(x) = \begin{cases} x^2 \cos \frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin \frac{1}{x}, & x > 0 \end{cases}$, then which of the following is (are) correct ?
- (A) $f(x)$ is continuous but not differentiable at $x = 0$
- (B) $f(x)$ is continuous and differentiable at $x = 0$
- (C) $f'(x)$ is continuous but not differentiable at $x = 0$
- (D) $f'(x)$ is discontinuous at $x = 0$

[MATCH THE COLUMN]

- | 11. | Column - I | Column - II |
|-----|--|-------------|
| (A) | If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$,
then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ equals | (P) 0 |
| (B) | Let $f(x)$ be a function satisfying the condition
$f(-x) = f(x)$ for all real x . If $f'(0)$ exists, then its
value is equal to | (Q) 1 |
| (C) | For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
the derivative from the left $f'(0-)$ equals | (R) 2 |
| (D) | The number of points at which the function
$f(x) = \max. \{a - x, a + x, b\}, -\infty < x < \infty,$
$0 < a < b$ cannot be differentiable is | (S) 3 |

EXERCISE # 6

1. Let $f(x)$ be defined in the interval $[-2, 2]$ such that $f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$ & $g(x) = f(|x|) + |f(x)|$.
Test the differentiability of $g(x)$ in $(-2, 2)$.
2. Discuss the continuity & the derivability in $[0, 2]$ of $f(x) = \begin{cases} |2x-3|[x] & \text{for } x \geq 1 \\ \sin \frac{\pi x}{2} & \text{for } x < 1 \end{cases}$
where $[.]$ denotes the greatest integer function
3. Examine the function, $f(x) = x \cdot \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ ($a > 0$) and $f(0) = 0$ for continuity and existence of the derivative at the origin.
4. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-3, 3]$ by $f(x) = \begin{cases} -x - [-x] & \text{if } [x] \text{ is even} \\ x - [x] & \text{if } [x] \text{ is odd} \end{cases}$
If L denotes the number of point of discontinuity and M denotes the number of points of non derivability of $f(x)$, then find $(L + M)$.
5. $f(x) = \begin{cases} 1-x & , (0 \leq x \leq 1) \\ x+2 & , (1 < x < 2) \\ 4-x & , (2 \leq x \leq 4) \end{cases}$. Discuss the continuity & differentiability of $y = f[f(x)]$ for $0 \leq x \leq 4$.
6. A derivable function $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ln(x/y) + x - y$ for every $x, y \in \mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.
7. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x = 0$, then find the value of $(n - 10m)$
8. If f is a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y) + f(0)}{3}$, $\forall x, y \in \mathbb{R}$ and $f'(0) = 2$, find $f'(x)$
9. If $\lim_{x \rightarrow 0} \frac{f(3 - \sin x) - f(3 + x)}{x} = 8$, then $|f'(3)|$ is
10. Let $f(x)$ be a differentiable function such that $2f(x+y) + f(x-y) = 3f(x) + 3f(y) + 2xy \forall x, y \in \mathbb{R}$ & $f'(0) = 0$, then $f(10) + f'(10)$ is equal to

ANSWER KEY

LIMIT

EXERCISE-1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. D | 4. C | 5. D | 6. B | 7. B |
| 8. C | 9. B | 10. B | 11. A | 12. A | 13. C | 14. B |
| 15. A | 16. C | 17. D | 18. A | 19. A | 20. C | 21. D |
| 22. D | 23. C | 24. B | 25. D | 26. A | 27. A | 28. A |
| 29. D | 30. D | 31. C | 32. D | 33. B | 34. C | 35. A |
| 36. D | 37. A | 38. C | 39. B | 40. C | 41. D | 42. A |
| 43. C | | | | | | |

EXERCISE-2

- | | | | | | |
|--|--------------------------|--------------|--|---------------------------|-------------------------|
| 1. 2 | 2. 5050 | 3. 2 | 4. $\frac{1}{32}$ | 5. $\frac{1}{16\sqrt{2}}$ | 6. $\frac{\sqrt{3}}{2}$ |
| 7. $\frac{1}{2}$ | 8. -2 | 9. $\pi - 3$ | 10. (i) $a = 1, b = -1$ (ii) $a = -1, b = \frac{1}{2}$ | | |
| 11. 1 | 12. $8\sqrt{2}(\ln 3)^2$ | | | | |
| 13. (a) $\pi/2$ if $a > 0$; 0 if $a = 0$ and $-\pi/2$ if $a < 0$; (b) $f(x) = x $ | | | | | |
| 14. -1/2 | 15. 16 | | | | |
| 16. e^{-8} | 17. $c = \ln 2$ | 18. e^{-1} | 19. $e^{-1/2}$ | 20. $\frac{\sqrt{3}}{2}$ | |

EXERCISE-3 (JM)

- | | | | | | | |
|-----------|-------|-------|-------|-------|-------|-------|
| 1. 1 | 2. 3 | 3. 1 | 4. 4 | 5. 1 | 6. 4 | 7. 3 |
| 8. 4 | 9. 1 | 10. 1 | 11. 1 | 12. 1 | 13. 1 | 14. 2 |
| 15. 3 | 16. 3 | 17. 4 | 18. 1 | 19. 2 | 20. 3 | |
| 21. 36.00 | 22. 4 | | | | | |

EXERCISE-4 (JA)

SECTION-1

- | | | |
|------|------|------|
| 1. D | 2. B | 3. B |
|------|------|------|

SECTION-2

- | | | | | |
|-------|------|------|-------|-------|
| 4. AC | 5. 0 | 6. 7 | 7. AD | 8. AD |
|-------|------|------|-------|-------|

EXERCISE-5

1. BCD 2. ABD 3. ACD 4. AC 5. BC 6. BD 7. BCD
8. AB 9. AD 10. AC 11. BC 12. AD 13. ABCD 14. BC
15. AC 16. ABC 17. AD 18. BCD 19. ABCD 20. BCD
21. $(A) \rightarrow (P,R); (B) \rightarrow (P,R); (C) \rightarrow (Q); (D) \rightarrow (Q)$
22. $(A) \rightarrow (R); (B) \rightarrow (S); (C) \rightarrow (P); (D) \rightarrow (Q)$
23. $(A) \rightarrow (P,Q,R,S,T); (B) \rightarrow (P,R); (C) \rightarrow (P,R); (D) \rightarrow (S)$

EXERCISE-6

1. e^{-1} 2. $(a_1, a_2, a_3, \dots, a_n)$ 3. $e^{\frac{1}{2}}$ 4. 167 5. $1/2$ 6. 8
7. $T(x) = \frac{1}{2} \tan^2 \frac{x}{2} \cdot \sin x$ or $\tan \frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2}x - \frac{1}{2}\sin x$, limit = $\frac{3}{2}$
8. $g(x) = \sin x$ and $\ell = e$ 9. $\frac{\theta}{\tan \theta}$ 10. $-\frac{1}{2e}$ 11. $a = -5/2, b = -3/2$
13. $\frac{2L}{3}$ 14. (a) 2, (b) D.N.E., (c) 0, (d) 0 15. (a) 2 ; (b) $1/2$

CONTINUITY

EXERCISE-1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. D | 4. A | 5. D | 6. C | 7. D |
| 8. B | 9. B | 10. A | 11. B | 12. D | 13. C | 14. B |
| 15. B | 16. D | 17. C | 18. D | 19. B | 20. B | 21. C |
| 22. A | 23. C | 24. A | | | | |

EXERCISE-2

1. -1 2. $a = 0, b = 1$ 3. $a = 0; b = -1$ 4. (a) -2, 2, 3; (b) $K = 5$; (c) even
5. P not possible. 6. (a) $4 - 3\sqrt{2} + a$, (b) $a = 3$
7. $g(x) = 2 + x$ for $0 \leq x \leq 1$, $2 - x$ for $1 < x \leq 2$, $4 - x$ for $2 < x \leq 3$, g is discontinuous at $x = 1$ & $x = 2$
8. $f(0^+) = -2$; $f(0^-) = 2$ hence $f(0)$ not possible to define
9. $a = 1/2, b = 4$ 10. $a = -3/2, b \neq 0, c = 1/2$

EXERCISE-3 (JM)

- | | | | | | | |
|------|------|-------|-------|---------|-------|-------|
| 1. 4 | 2. 1 | 3. 4 | 4. 3 | 5. 1 | 6. 4 | 7. 2 |
| 8. 2 | 9. 3 | 10. 4 | 11. 3 | 12. [5] | 13. 2 | 14. 4 |

EXERCISE-4 (JA)

SECTION-1

1. BD 2. AD 3. ACD

SECTION-2

4. Discontinuous at $x = 1$; $f(1^+) = 1$ and $f(1^-) = -1$

EXERCISE-5

1. ABCD 2. AB 3. ABC 4. BCD 5. ABCD 6. ACD 7. ACD
8. ACD 9. ABCD 10. CD
11. $(A) \rightarrow (Q); (B) \rightarrow (R); (C) \rightarrow (S); (D) \rightarrow (P)$
12. $(A) \rightarrow (P, Q, R); (B) \rightarrow (P, Q, T); (C) \rightarrow (P, Q, T); (D) \rightarrow (P, Q, S)$

EXERCISE-6

1. $A = -4, B = 5, f(0) = 1$
2. $f(0^+) = \frac{\pi}{2}; f(0^-) = \frac{\pi}{4\sqrt{2}} \Rightarrow f$ is discount. at $x = 0$; $g(0^+) = g(0^-) = g(0) = \pi/2 \Rightarrow g$ is cont. at $x = 0$
3. discontinuous at all integral values in $[-2, 2]$
4. locus $(a, b) \rightarrow x, y$ is $y = x - 3$ excluding the points where $y = 3$ intersects it.
5. $c = 1, a, b \in \mathbb{R}$ 6. $e^2 + e^{-2}$
7. $k = 0; g(x) = \begin{cases} \ln(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} \leq x < \frac{\pi}{2} \end{cases}$. Hence $g(x)$ is continuous everywhere.
8. $g(x) = 4(x + 1)$ and limit $= -\frac{39}{4}$ 9. (a) 5 (b) 30

DIFFERENTIABILITY

EXERCISE-1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. C | 3. A | 4. A | 5. C | 6. B | 7. D |
| 8. A | 9. B | 10. D | 11. B | 12. B | 13. A | 14. D |
| 15. D | 16. A | 17. D | 18. D | 19. D | 20. C | 21. D |
| 22. D | 23. A | 24. D | | | | |

EXERCISE-2

1. $f(x)$ is conti. but not derivable at $x = 0$
2. conti. $\forall x \in \mathbb{R}$, not diff. at $x = 0, 1$ & 2
3. $0 < n \leq 1$
4. conti. but not diff. at $x = 0$; diff. & conti. at $x = \pi/2$
5. conti. but not diff. at $x = 0$
7. f is conti. but not diff. at $x = 0$
8. $f'(1^+) = 3, f'(1^-) = -1$
9. $a = 1/2, b = 3/2$
10. 3

EXERCISE-3 (JM)

- | | | | | | | |
|-------|------------|-------|-------|-------|-------|-------|
| 1. 2 | 2. 1 | 3. 4 | 4. 4 | 5. 2 | 6. 3 | 7. 3 |
| 8. 1 | 9. 4 | 10. 4 | 11. 2 | 12. 2 | 13. 3 | 14. 4 |
| 15. 1 | 16. [3.00] | | | | | |

EXERCISE-4 (JA)

SECTION-1

1. C 2. B

SECTION-2

3. AC 4. BC 5. ABCD 6. D 7. 3 8. AB 9. BC
10. 2.00 11. D

EXERCISE-5

1. ABC 2. BC 3. ABC 4. BC 5. ABCD 6. ABCD
7. BCD 8. BD 9. ABD 10. BD 11. (A) R, (B) P, (C) Q, (D) R

EXERCISE-6

1. not derivable at $x = 0$ & $x = 1$
2. f is conti. at $x = 1, 3/2$ & disconti. at $x = 2$, f is not diff. at $x = 1, 3/2, 2$
3. If $a \in (0, 1)$ $f'(0^+) = -1$; $f'(0^-) = 1 \Rightarrow$ continuous but not derivable
 $a = 1$; $f(x) = 0$ which is constant $\Rightarrow$ continuous and derivable
 If $a > 1$ $f'(0^-) = -1$; $f'(0^+) = 1 \Rightarrow$ continuous but not derivable
4. 8
5. f is conti. but not diff. at $x = 1$, disconti. at $x = 2$ & $x = 3$. cont. & diff. at all other points
6. 5150 7. 74 8. $f(x) = 2x + c$ 9. 4 10. 120

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018

AIR
29



Sarthak Behera

AIR
36



Shashank Agrawal

AIR
46



Shreya Pathak

AIR
48



Raaghav Raaj

AIR
2
SC



Aayush Kadam

AIR
18
OBC



Gunjan

Result Highlights 2017

AIR
12



Abhay Goyal

AIR
14



Piyush Tibarwal

AIR
20



Abhinav Kumar

AIR
38



Tushar Gautam

AIR
49



Saurav Sahoo

AIR
60



Pratyush

Mentored by Team

AIR
4



Kushal Babel
2014

AIR
15



Gourika Bang
2015

AIR
21



Amit Malav
2013

AIR
24



Suraj Sanjay Jog
2012

AIR
38



Chitraang Murdia
2014

AIR
58



Pulkit Tandon
2012

AIR
68



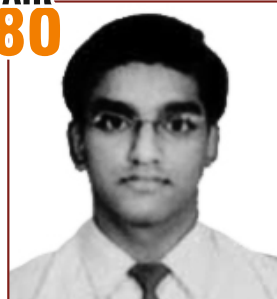
Kapil Shobhnani
2016

AIR
70



Utkarsh Sinha
2015

AIR
80



S John S
2015